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RTD Denver EMU

Project # 7230

Portable Test Unit Application

Software Integration Test Plan (SITP)

Contract Number	-	CDRL Number	-
		Customer Ref. Number	-

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Preface

This Software Integration Test Plan (SITP) is prepared in accordance with IEEE-829-1998, with slight adaptation to the format. It has been adapted to combine the 3 documents: Test Plan, Test Design Specification, and the Test Procedure Specification.

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1 TEST PLAN SPECIFICATION IDENTIFIER

201111464 Revision 1, 2011-11-04

2 INTRODUCTION

This document is the TC3 Portable Test Unit (PTU) Application Software Integration Test Plan (SITP) developed by Vapor Stone Rail Systems.

This document includes 3 major contents as stated in Software Quality Assurance Plan (SQAP) P/N 201111208, Section 4.9:

- Section 3 Test Plan:
  Identifies the tests required to verify and validate the PTU Aplication. Its content complies with the requirements of IEEE STD 829, standard for Software Test Documentation.

- Section 4 Test Design:
  Describes the design of the tests to be performed in order to validate the requirements of the PTU Aplication.

- Section 5 Test Procedure:
  Describes the procedures that need to be executed in order to validate the requirements of the PTU Aplication.

2.1 Objectives

The objective is to plan the way the Software Requirements Specification (SRS) will be tested as well as to define the tests required to validate Portable Test Unit Application.

2.2 Scope

The scope of this SITP applies to the software requirements of the PTU Application.

The intended audience is the Vapor Stone Rail Systems software development and maintenance team members of the Hyundai Rotem and the Denver Regional Transportation District.

2.3 Definitions and Acronyms

2.3.1 Definitions

Boot Loader:	Startup software verifies the memory and integrity of the application before jumping to the main application.
Door Control Unit:	A processor based electronic unit running software that controls the door, provides monitoring functions, and interacts with the user via a PTU Application and a network.
Graphical User Interface:	Interface used for the communication between the User and the Portable Test Application software. This is the visible part of the Portable Test Application
User:	A human operating the Portable Test Application, issuing request and reading results

2.3.2 Acronyms

Acronym	Meaning
CL	Car Line
CPLD	Complex Programmable Logic Device
CPU	Central Process Unit
CRC	Cyclic Redundancy Check
DCML	Door Cut-out Mechanical Lock
DCU	Door Control Unit
EDF	Event Description File
EMU	Electric Multiple Units
I/O	Input/Output
ICD	Interface Control Document
ID	Identification
IEEE	Institute of Electrical and Electronics Engineers
LED	Light-Emitting Diode
OC	Over Current
OPDF	Operating Parameters Description File
P	Pulse
P/N	Part Number
PC	Personal Computer
PTU	Portable Test Unit
PWM	Pulse Width Modulation
PWR	Power
RAM	Random Access Memory
RTD	Regional Transport District
RTN	Return
RTP	Routine Test Procedure
RX	Receive
S/R	Set/Reset
SACS	Software Applications Configuration Specifications
SCI	Software Configuration Item
SDD	Software Design Description
SITP	System Integration Test Plan
SITR	Software Integration Test Report
SPR	Software Problem Report
SQAP	Software Quality Assurance Plan
SRAM	Static Random Access Memory
SRS	Software Requirements Specifications
SRTM	Software Requirements Traceability Matrix

Acronym	Meaning
SVVP	Software Verification and Validation Plan
TBD	To Be Defined
TL	Train Line
TS	Technical Specifications
TX	Transmit
VSRS	Vapor Stone Rail Systems

2.3.3 References

This section lists the number, title, revision, date, and source of all documents referenced in this Software Integration Test Plan.

Standards

IEEE Std 829-1998	IEEE Standard for Software Test Documentation, September 1998, IEEE, www.ieee.org
IEEE Std 1558-2004	IEEE Recommended Practice for Software Documentation for Rail Equipment and Systems, February 2nd 2005, IEEE, www.ieee.org

Vapor Stone Rail Systems Documents

201111185	System Functional Description, RTD Denver EMU Door System, Rev. 3
201111186	Software Configuration Management Plan, Door System, Rev. 1
201111207	Software Project Management Plan, Door System, Rev. 0
201111208	Software Quality Assurance Plan, Door System, Rev. 1
201111254	Software Verification and Validation Plan, Door System, Rev. 0
201111274	Generic Software Requirement Specification, Door System -Portable Test Application, Rev. 1.
201111295	Generic Software Design Description, Door System - Portable Test Application, Rev. 1
201111448	Software Requirements Traceability Matrix (SRTM), Rev. 01
201111489	Generic Software Documentation Project – Portable Test Interface Protocol, Interface Control Document (ICD), Rev. 01
201111576	Software Applications Configuration Specifications (SACS), Denver Door System, Rev. 01

VSRS Test Report Document

201111465	Software Integration Test Report, Portable Test Unit Application, Rev. TBD.

3 TEST PLAN

3.1 Test Items

All the features that make up the Portable Test Unit Application will be tested during the test phase. The version tested will be placed under configuration management.
The Software Requirement Specification (SRS) of the Portable Test Unit Application and Software Applications Configuration Specifications (SACS) document will provide the basis for defining the correct operation.
Table 3-1 lists the item to be tested.

Table 3-1: Item to Be Tested

Item	Software #
Portable Test Unit Application	2011L0054

3.2 Features to be tested

The following features of the PTU available for the DCU Boot Loader will be tested with the PTU simulator, a software application simulating the DCU Boot Loader.

- Valid Login and Logout
- Invalid Login/Physical Disconnection
- Valid Command (DCU Boot Loader Protocol)
- Valid File Transfer (DCU Boot Loader Protocol)
- Retry Testing File Transfer (DCU Boot Loader Protocol)
- Data Communication Error Detection and Process
- Request the Status Code of the DCU Boot Loader
- Software Download (DCU Application)
- Download Static RAM Content
- Security Check
- PTU User Interface (Boot Loader Screens)
- PTU Connection to Boot Loader Mode
- Display Window Guidelines (Boot Loader Screens)

The following features of the PTU available for the DCU Application will be tested with the PTU simulator, a software application simulating the DCU Application.

- Valid Login and Logout
- Invalid Login/Physical Disconnection
- Valid Command (DCU Application Protocol)
- Valid File Transfer (DCU Boot Loader Protocol)
- Retry Testing File Transfer (DCU Application Protocol)
- Data Communication Error Detection and Process
- Request the Status Code of the DCU Application
- Start, Stop and Display Real-Time Monitoring
- Log File Download and Delete
- Snapshot File Download and Delete
- Door Movement Information Graph Download and Delete
- Self-Test Results Display
- Send and Receive Door Configuration
- Read and Display Door Identification
- Download Static RAM Content
- EDF and OPDF Download
- Retrieve and Display all Software Versions Used in the DCU
- Security Check
- PTU Connection to DCU Application
- File Structure

- Device Status Representation
- Local Equipment Date and Time Update
- DCU System Counters and its resetting

The following features of the PTU will be tested with the PTU.

- Program Installation and Removal
- PTU Start and Exit
- Open and Save of Data File
- Modified Working Directory
- Display Help Information
- Print Selected Data

The following features of the PTU will be tested with the PTU connected to the DCU.

- Software Download (DCUs, Boot Loader and Neuron Gateway Application)

The following characteristics of the PTU will be verified during testing:

- PTU User Interface (General screens, Boot Loader and DCU Application Screens)
- Display Window Guidelines (General screens, Boot Loader and DCU Application Screens)

The serial port setting modification feature of the PTU will be tested using the PTU Simulator simulating the DCU Boot Loader. Port settings will be changed on the PTU and the PTU Simulator and a login will be performed. All the available port settings will be tested.

3.3 Features not to be tested

All the features of the Portable Test Unit Application are tested.

3.4 Approach

The test personnel will use the SFD, SRS, SACS, Portable Test ICD, and SDD to prepare all test design, case, and procedures specifications. This approach will verify the accuracy and comprehensiveness of the information in the documentation in those areas covered by the tests.

The test personnel will run the test procedures and complete the test reports. The test data are to be under configuration management.

The following software programs will be used in the test procedure and if they don't exist, they will need to be developed prior to executing the test procedure.

- Portable Test Unit Simulator (PTU Simulator)
- Boot Loader application
- Door Control Unit (DCU) application

The PTU Simulator runs on a PC and communicates using the serial port. This application allows sending and receiving commands (valid and invalid) and data. This will help testing error handling.

The Software Requirements Traceability Matrix (SRTM) will be used to trace all requirements from the SRS to the related test cases "Results" section.

3.5 Item Pass/Fail Criteria

The Portable Test Unit Application software is to be considered acceptable if:

1. It meets all the criteria defined in the Software Requirements Specifications (SRS) – Portable Test Unit Application, SACS document or upon approval of non-conformities detected during the test by the customer.
2. After undergoing all the tests, the Portable Test Unit Application software remains functional without corruption of settings

3.6 Suspension Criteria and Resumption Requirements

3.6.1 Suspension Criteria

The following criteria will cause a suspension of all testing activities:

- Failure of one of the test cases.

3.6.2 Resumption Requirements

When a new version of the Portable Test Unit Application software is transmitted to the test personnel after a suspension of testing has occurred, the procedure will be started again if more than 10% of the tests lead to failure. If the failure percentage is lower, regression testing can be used.

3.7 Test Deliverables

The following documents will be generated by the test personnel and will be placed under configuration management:

- This document, PTU Application SITP
- PTU Application Software Integration Test Report (SITR)

3.8 Testing Tasks

The testing tasks are defined in the SVVP section 5 (V&V Processes)

3.9 Environmental Needs

3.9.1 Hardware

The materials and equipment used for this test plan are those called in the specific test procedures and test cases. The hardware used for the test is listed in the table Table 5-1.

3.9.2 Software

All software applications used are those called in the specific test design and test procedure. The software used for the test is listed in the Table 5-2.

3.10 Responsibilities

The responsibilities are defined in the SVVP section 4.5.

3.11 Staffing and Training Needs

The staffing needs are defined in the SVVP section 4.6. No training is needed for this project.

3.12 Schedule

The schedule is defined in the SVVP section 4.2.

3.13 Risks and Contingencies

The risks and contingencies are defined in the SQAP section 14.

3.14 Approvals

Approvals' names and signatures are included in the cover page of this document. Results approval are approved as a separate document #201111465 PTU Application SITR.

4 TEST DESIGN

4.1 Test Identification

The Table 4-1 lists the test identification for this document.

Table 4-1: Test Identification

#	Test Identification	Test Description
1	Software Installation	• Verify the PTU Software provided on CD-ROM. • Verify if the PTU Application Installation Package contains all necessary files to install the PTU and it's sub components.. • Verify the PTU installation. • Verify if the PTU Application executable file has a unique name “PTEProgram.exe”. • Verify if the program exits from “Exit” option on the “File” menu from the menu bar.
2	Graphical User Interface	• Verify the PTU Application Graphical User Inerface.
3	PTU Application Properties	• Verify the Serial Port setting read at start-up. • Verify if the Serial Port settings can be modified. • Verify the Serial Port setting . • Verify if the Serial Port setting error is reported to user. • Verify the communication speed.
4	Login/Logout	• Verify if the PTU Application logs in automatically with the DCU when launched. • Verify DCU Boot Loader valid login • Verify DCU Boot Loader valid command testing • Verify DCU Boot Loader valid logout • Verify DCU Boot Loader invalid login • Verify physical disconnection from DCU after login to DCU Boot Loader • Verify PTU exit after login to DCU Boot Loader • Verify DCU Application valid login • Verify DCU Application valid command testing • Verify DCU Application valid logout • Verify DCU Application invalid login • Verify physical disconnection from DCU after login to DCU Application • Verify PTU exit after login to DCU Application • Verify PTU login option
5	Communication Error Process	• Verify DCU to PTU CRC error report • Verify communication time-out triggering retries • Verify communication error after 3 tries
6	EDF and OPDF	• Verify EDF update and browse • Verify EDF download • Verify OPDF update and browse • Verify OPDF download
7	Request DCU Code Status	• Verify the Code Status of the DCU Boot Loader • Verify the Code Status of the DCU Application • Verify the Code Status of the DCU Neuron Application
8	Read Location Identification and Car Number	• Verify Location ID request • Verify Location ID received • Verify Location ID displayed on the left side control menu when PTU Application starts • Verify Car Number displayed on the left side control menu when PTU Application starts

#	Test Identification	Test Description
9	Software Versions	• Verify versions request from PTU • Verify versions retrieved by PTU • Verify the following versions display: DCU Application, DCU Boot Loader, DCU Neuron Application, and CPLD.
10	Real-Time Monitoring	• Verify the start of the real-time data monitoring request • Verify real-time data monitoring • Verify the stop of the real-time data monitoring request
11	Retrieve Self Test	• Verify Self Test results request • Verify Self Test results display
12	DCU Status	• Verify DCU Status request • Verify DCU Status displayed information • Verify event synchronization • Verify that the system status and functional information are provided in a graphical format, superimposed over a system schematic or diagrammatic. When a graphic presentation is not practical, system status and functional information are provided in English language statements and relevant engineering units, and are formatted for simple readability. • Verify that the system status and functional information conform to the following principles: “- In the case of units of temperature, absolute units (Kelvin, Rankin) shall not be used, and prefixing the For C symbol with a small circle (degree symbol) shall not be used: instead, "deg" or "degrees" shall be the prefix, e.g. "deg F", "degrees C" etc.”
13	Door Open Time Configuration	• Verify Door Open Time Configuration from DCU • Validate Door Open Time Configuration entered by user • Verify sending Door Open Time Configuration to DCU • Verify Door Open Time Configuration reception confirmation from DCU
14	Door Close Time Configuration	• Verify Door Close Time Configuration from DCU • Validate Door Close Time Configuration entered by user • Verify sending Door Close Time Configuration to DCU • Verify Door Close Time Configuration reception confirmation from DCU
15	Obstruction Retries Times Configuration	• Verify Obstruction Retries Times Configuration from DCU • Validate Obstruction Retries Times Configuration • Verify sending Obstruction Retries Times Configuration to DCU • Verify Obstruction Retries Times Configuration reception confirmation from DCU
16	Obstruction Re-Open Distance Configuration	• Verify Obstruction Re-Open Distance Configuration from DCU • Validate Obstruction Re-Open Distance Configuration • Verify sending Obstruction Re-Open Distance Configuration to DCU • Verify Obstruction Re-Open Distance Configuration reception confirmation from DCU
17	Delay Before Re-closing Configuration	• Verify Delay before Re-closing Configuration from DCU • Validate Delay before Re-closing Configuration • Verify sending Delay before Re-closing Configuration to DCU • Verify Delay before Re-closing Configuration reception confirmation from DCU
18	Delay Before Door Closing Configuration	• Verify Delay Before Door Closing Configuration from DCU • Validate Delay Before Door Closing Configuration • Verify sending Delay Before Door Closing Configuration to DCU • Verify Delay Before Door Closing Configuration reception confirmation from DCU

#	Test Identification	Test Description
19	Door Open Duration for Passenger Door Enable Configuration	• Verify Door Open Duration for Passenger Door Enable Configuration from DCU • Validate Door Open Duration for Passenger Door Enable Configuration • Verify sending Door Open Duration for Passenger Door Enable Configuration to DCU • Verify Door Open Duration for Passenger Door Enable Configuration reception confirmation from DCU
20	Time of the DCU	• Verify DCU time request • Verify display time
21	Download Memory Content	• Verify downloading SRAM content request • Verify downloading SRAM
22	Door Movement Information Graph	• Verify downloading door graph request • Verify door graph download • Verify door graph display
23	Log File and Snapshots	• Verify log file download request • Verify log file download • Verify log file storing and display • Verify log database sorting • Verify log save • Verify log load • Verify log file download communication error • Verify log file delete request • Verify log file delete confirmation asked • Verify log file delete confirmation from DCU. • Verify snapshot file download request • Verify snapshot file download • Verify snapshot file display • Verify snapshot file download communication error • Verify snapshot file delete request • Verify snapshot file delete confirmation by user • Verify snapshot file delete confirmation from DCU.
24	Data and File Management	• Verify save data file • Verify open data file • Verify display data file • Verify disk error while saving data file • Verify invalid data file name
25	Upgrade of DCU Application Executable Code	• Verify application upgrade request • Verify application upgrade file transfer • Verify application upgrade process monitoring • Verify application upgrade interrupted • Verify application upgrade communication error • Verify physical disconnection from DCU during application upgrading
26	Upgrade of DCU Boot Loader	• Verify DCU Boot Loader upgrade request • Verify RTP Tool upgrade request
27	Upgrade of DCU Neuron Application	• Verify DCU Neuron Application upgrade request
28	PTU Security	• Verify PTU Access level changes with correct password • Verify DCU Application Upgrade, Flash Programming, Neuron Application Upgrade, DCU Configuration, Snapshot Erasure, DCU Log and System Counters resetting are available when PTU Access is administrator
29	Help File	• Verify Display Help Information
30	Printing	• Verify Print Selected Log and Snapshot Files

#	Test Identification	Test Description
31	System Counters	• Verify Request DCU System Counters and Display • Verify System Counters Download • Verify the System Counter Reset
32	Program Removal	• Verify if Uninstall Program is registered so that it is listed on the Uninstall page of the Add/Remove Program Windows Utility. • Verify PTU Software Removal • Verify if Uninstall Program removes all files in Program Files folder that it uniquely owns. • Verify if Uninstall Program leaves no remaining file or registry entry other than the files created by the user.
33	File Structure and Security Restrictions	• Verify if PTU Application works for any drive letter (C:, D:, …) • Verify if all PTU Application generated data are stored in X:\DENVERPTU\DATA\DOOR • Verify if “Everyone” has read / write access to X:\DENVERPTU\DATA\DOOR • Verify if “Everyone” has read access to X:\DENVERPTU\Program\DOOR-Door Subsystem\DOOR-x.y.z • Verify if executable program, help file, DLL and any other data that allow PTU Application to work are stored in X:\DENVERPTU\Program\DOOR-Door Subsystem\DOOR-version-note • Verify if software download packages are located in X:\DENVERPTU\SWDownload\DOOR • Verify if temporary files are stored in X:\DENVERPTU\Temp and are deleted when PTU application exits
34	Date and Time Presentation	• Verify if all dates displayed by the PTU use the form MM-DD-YYYY. Month, day and year use Integer values. Year uses four digits • Verify if all times displayed by the PTU use the form HH:MM:SS. “HH” is hours from 00 to 23, “MM” is minutes from 00 to 59, and “SS” is seconds from 00 to 59 • Verify if the PTU Application is able to process all dates in the range of 2000 to 2050

5 TEST PROCEDURE

5.1 Special Requirements

The following tables list the equipment, the software and the special files used in the testing:

Table 5-1: Required Hardware

Qty	Description	Version	P/N	Revision
2	IBM compatible computer running under Windows 2000/XP	N/A	N/A	N/A
1	Null Modem Cable	N/A	TBD	0
1	PTU RS-232 Cable	N/A	TBD	0
1	Printer with cable or network access	N/A	N/A	N/A
1	DCU Assembly	---	TBD	---
1	Variable Power Supply (Voltage source min. 10A, 0-50V)	N/A	---	---
1	Bootstrap/Reset Box	N/A	---	---

Table 5-2: Required Software

Description	Version	P/N	Revision
PTU installation package generates: • PTU software • .NET Framework	0.0.0 0.0.0 3.5	2011L0054 N/A N/A	--- --- N/A
PTU Simulator Software	0.0.0	N/A	---
DCU Application Software	--	2011L0051	---
DCU Boot Loader Application Software	--	2011L0053	---
Neuron Gateway Application	--	2011L0052	---

5.2 Procedure Steps

5.2.1 Log

A Software Integration Test Report will be generated to record the test results based on this document
The result of each test case log should be recorded in the last column, “P/F”,(see Section 5.3). Letter “P” or “F” will indicate Pass or Fail to the test.
A Test Report Sheet included in Appendix B of this document will be correctly filled (including Software Problem Report (SPR) if any) and dated.
The tool used for SPR tracking is the ClearQuest database.

5.2.2 Test Setup

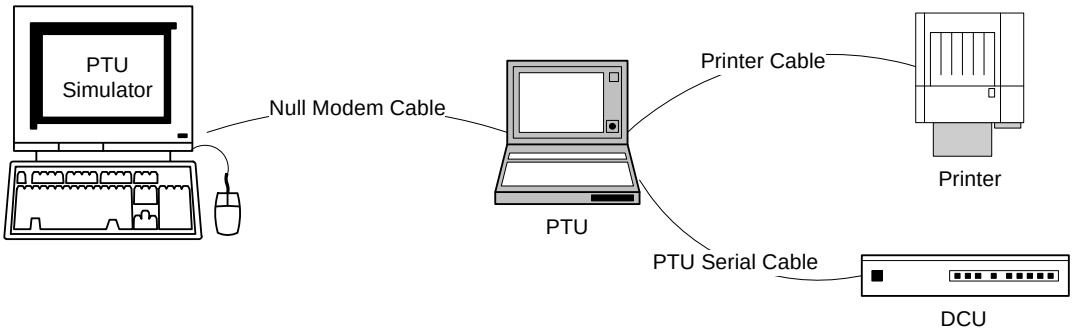

Figure 5-1: Test Setup

5.2.2.1 PTU Application Installation

The PTU software comes with an automated Windows-based installation program called SETUP. Don't try copying the files manually on your hard disk; instead please follow these steps to install the software:

1. Extract from the PTU installation zip file 2011L0036.ZIP the setup.exe file that contains installation for the latest version of the PTU. A good practice would be to stop any other application before running setup.
2. Run SETUP.EXE.
3. The setup program will then begin to copy and decompress files on the target drive.
4. The setup program will prompt you to terminate any application that might lock system resources or files.
5. The next step will be to select a target directory for the program. It is recommended that the default location will be used if possible.
6. Once the target directory has been selected and approved, the setup program initiates the installation procedure. Once the process is completed, the program will inform you if the installation was successful or not.

5.2.2.2 PTU Simulator

5.2.2.2.1 PTU Simulator Installation

The PTU Simulator software comes with an automated Windows-based installation program called SETUP. Don't try copying the files manually on your hard disk; instead please follow these steps to install the software:

1. Extract from the PTU Simulator installation zip file.
    - Note: A good practice would be to stop any other application before running setup.
2. Run SETUP.EXE.
3. The setup program will then begin to copy and decompress files on the target drive.
4. The setup program will prompt you to terminate any application that might lock system resources or files.
5. The next step will be to select a target directory for the program. It is recommended that the default location will be used if possible.
6. Once the target directory has been selected and approved, the setup program initiates the installation procedure. Once the process is over, the program will inform you if the installation was successful or not.

5.2.2.2.2 PTU Simulator Description

The PTU Simulator Software is used to simulate all the DCU Application and DCU Boot Loader commands. It is to validate the PTU response to the DCU Application and DCU Boot Loader. The PTU Simulator and the PTU are installed on a personal computer with Windows XP.

There are seven data simulation files included in the PTU Simulator:

- event.log: event log simulation file.
- snapshot.lss: snapshot simulation file.
- memory.bin: DCU memory simulation file.
- graph.bin: DCU door graph simulation file.
- systemCounters.bin: System counters simulation file.
- EDF.bin: Event Description file.
- OPDF.bin: Operating Parameter Description file.

The PTU Simulator is shown on the following figure:

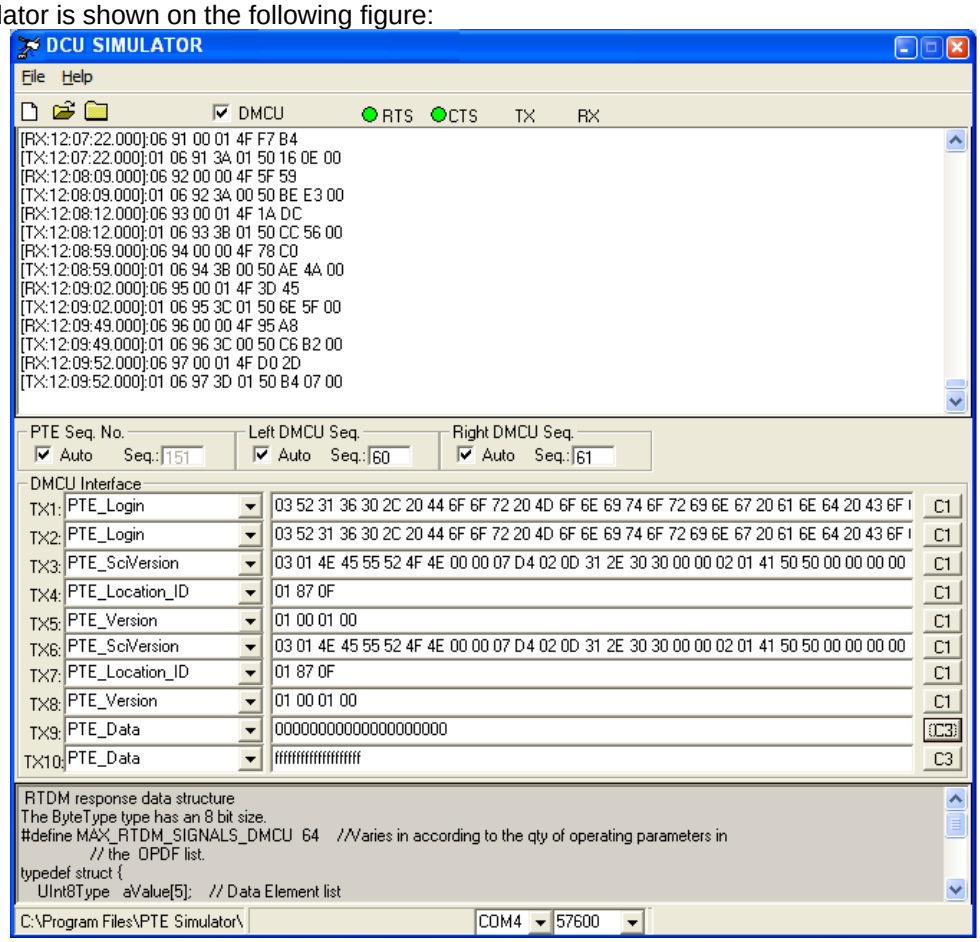

Figure 5-2: PTU Simulator User Interface

The PTU Simulator use is as follows:

- To perform PTU test, the DCU check-box must be checked to let the PTU Simulator simulate all the DCU Application & DCU Boot Loader functions.
- The PTU Simulator will automatically respond to the file download/upload request.
- The PTU Simulator user can send data via 3 channels manually:
  - C1: This channel is used to send general PTU request response except the file transfer (download & upload) command, and real time monitoring command.
  - C2: This channel is used to send real time event data.
  - C3: This channel is used to send real time data.
- To change the channel, right-click the channel button, the channel name will be changed from C1->C2->C3->C1… so on.
- To enable sending data from channel upon reception of request, left-click the channel button and the channel button is changed to green. The channel button is changed back to normal (gray) after the data is sent.
- The PTU Simulator will pick up one data from channel automatically and send to PTU upon reception of PTU request. The data pick up sequence will be from TX1 to TX10.

5.2.2.3 Boot Loader Programming

To upload the Boot Loader Application in the target module, you have to execute the following steps:

1. Turn DCU power OFF.
2. Connect PTU computer and DCU with PTU RS-232 cable.
3. On the PTU RS-232 cable, turn the switch to “Boot Strap” mode.

4. On the PTU Application, unlock administration functions: go to menu Administration → Other Functions → Password, write **vsrsdenver** and press OK.
5. Select Tools → Load Flash → Open Flash File to open the Boot Loader Application.
6. Select Tools → Load Flash → Send Flash File to start to upload the file. Follow the uploading prompt of the PTU (turning on the DCU and pressing "OK" to PTU Application message).
    - Note: When DCU boots up correctly in bootstrap mode, three LED (Running, Warning, and Fault) are all turned on.
7. After Boot Loader Application upload process is finished, verify if the uploading successful information is presented. If the uploading is successful, follow the prompt to remove the PTU cable (or turn the switch to DCU mode) and reset the DCU. If the uploading is failed, restart from step 1.
8. The target module is programmed with the Boot Loader Application version.

5.2.2.4 DCU Application Programming

To upload the DCU Application into the DCU, you have to execute the following steps:

1. Connect PTU computer and DCU with PTU RS-232 cable.
    - Note: Make sure the switch is not in Boot Strap mode in PTU cable.
2. Turn DCU power ON.
3. Start PTU.
4. To unlock administration functions, go to menu Administration → Other Functions → Password, write **vsrsdenver** and press OK.
5. Select Tools → DCU Upgrade → Open DCU Application File.
6. Select DCU Application file and click OK
7. After the file verification, click OK
8. Select Tools → DCU Upgrade → Upgrade.
9. Click YES.
10. When the upload is finished successfully, click OK on the prompt window

5.2.3 Other Information about Test

In this test procedure, each test section is independent. Within a section, the initial setup is described in the "Special Procedural Requirements" and all test cases must be executed sequentially.

The results to be checked of each test case are defined in the Output Specification section. There are two generated logs to verify communications or background processes: the log window in the PTU Simulator and the debug and error log in the PTU. Using them, in addition to the GUI displays, will allow the tester to measure the functionality.

When timing measurements are required between communications, the PTU Simulator and the PTU logs contain time stamps. The PTU Simulator time stamps have precedence over the PTU and are used as a reference to calculate timings.

In uploading files, it is sometimes required to compare two files. The DOS command "FC.EXE" will be used to verify the binary file in the test procedure. This software is part of Windows OS and is called within a Command Prompt window (Start → Run…, and type "cmd").

All LEDs described in this test procedure are on the DCU.

Do all tests with the power supply set to nominal voltage.

All tests should be performed on one DCU by default, unless otherwise stated.
Any anomalies that would prevent the test case from being completed must be noted, on the page of the test in question. The test case result will be marked as Failed.

Each anomaly that is not due to a PTU problem must be brought up to the attention of the Software Team Leader. Appropriate action will be taken in accordance to the current situation.

5.3 Test Cases

It is recommended that the Test Cases inside a section will be done sequentially.

5.3.1 TC3.1 Software Installation

Special Procedural Requirements: If conflict of DLL files occurs during installation, choose to keep the newest ones.
Measurements: Self-explanatory.

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
1	TC3.1.1 PTU Software Provided on CD-ROM	Verify if the PTU installation package is provided on CD-ROM.	The PTU installation package is provided on CD-ROM.		
2	TC3.1.2 PTU Application Installation Package Contains All Necessary Files To Install the PTU and it's Sub Components	Open the .ZIP file and verify the file contents.	The installation package includes at least: -setup.exe -Readme.txt, containing the proper version number and instructions -The PTU installer (.msi) file -The .NET Framework 3.5 -The Echelon OpenLDV installer		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
3	TC3.1.3 PTU Application Installation	1) Make sure the PTU is not already installed by going to "C:\DenverPTU\Program\DOOR-Door Subsystem\DOOR-*x.y.z*", where *x.y.z* represents the version of the PTU under test. Start PTU installation package **setup.exe**. Install the PTU software in default directory (NOTE: The setup must automatically install the **.NET Library version 3.5** and the **OpenLDV Echelon's** driver if not already on the computer). 2) Verify if any error message appears during the installation. Verify if the "Denver PTU: VAPOR Door Setup was completed successfully" message pops up at the end of installation. Click "OK". 3) Verify that "PTUProgram.exe" file is found in the default installation directory (C:\ DenverPTU \Program\ Door Subsystem \DOOR-*x.y.z*). 4) Verify that PTU can be launched via Windows Start menu, and it has no shortcut on desktop.	1) The PTU installation package is started. 2) No error message appears. The "Denver PTU: VAPOR Door Setup was completed successfully" message pops up. 3) The "PTUPROGRAM.exe" file is found in default installation directory. 4) The PTU launches from Windows Start menu, and there are no shortcuts for it on desktop.		
4	TC3.1.4 Unique Command "PTEProgram.exe" Required to Start PTU Application	Go into the PTU software subdirectory "C:\ DenverPTU \Program\DOOR- Door Subsystem \DOOR-*x.y.z*". Create a shortcut of "PTUProgram.exe". Drag the shortcut to Windows desktop. Double click the shortcut to start PTU Application.	The PTU main window pops up.		
5	TC3.1.5 Program exit supported from "Exit" option on the "File" menu from the menu bar.	In PTU Application, select "Exit" from the "File" menu. Click "No" when asked if you want to save the project file.	The PTU application closes.		

5.3.2 TC3.34 Graphical User Interface

Special Procedural Requirements: None
Measurements: Self-explanatory.

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
1	TC3.34.1 PTU Application Graphical User Interface	Double click on the file "PTUProgram.exe" found in the default installation directory (C:\ DenverPTU \Program\ Door Subsystem \DOOR-*x.y.z*) Using the Figure A-3, verify if menus are present in the PTU Application.	PTU Application starts. In the PTU Application, the menus are present as in Figure A-3.		

5.3.3 TC3.2 PTU Application Properties

Special Procedural Requirements: Before testing, start the PTU Simulator on the PTU Simulator computer. Make sure that the serial port of the PTU Simulator is correctly configured. (The serial port configuration of the PTU Simulator is displayed on the PTU Simulator Status Bar. The Speed should be 115200 bps. The Port number should be the port that connects to the PTU. Click on the Status Bar to change the configuration.)
For serial port configuration of the PTU Application, only the serial port number is accessible by the user.
Measurements: Self-explanatory.

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
1	TC3.2.1 Serial Port Setting Read at Start-Up	Verify the PTU status bar.	"COM1" (or current setting) is displayed on the PTU status bar.		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
2	TC3.2.2 Modify Serial Port Settings	1) In the PTU File menu, select Properties. 2) In the Properties window Port text, change the serial port number. Change the Login Option to Boot Loader. Click OK button. 3) In the PTU File menu, click Exit to close PTU Application. 4) Start PTU Application again from PTU computer. 5) In the PTU File menu, select Properties. 6) Verify that the last modifications remain. 7) Change the communication serial port to the port that connects to PTU Simulator. Change the Login Option to DCU Application. Click “OK” to save settings. 8) Close PTU Application.	1) The PTU Properties window pops up with the last saved settings (the default values are: Port number is Com 1 and Login Option is DCU Application). 2) The PTU Properties window is closed. 3) PTU Application is closed. 4) The PTU main window pops up. 5) The PTU Properties window pops up. 6) The last modifications remain. 7) The PTU Properties window is closed. 8) The PTU main window is closed.		
3	TC3.2.3 Serial Port Setting	See [Modify serial port settings].			

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
4	TC3.2.4 Serial Port Setting Error Reported to User	1) On the PTU Computer, start a software application that uses the same Serial Port number as the PTU Application (e.g. PTU Simulator), then, start the PTU Application. 2) Click “OK”. 3) Click “Cancel” button on the Properties window. 4) Close the software application that uses the same Serial Port number as the PTU Application. In the PTU Application, select Properties from File menu. 5) Change the Serial Port number and the Login Option. Click “Cancel” to close properties window. 6) Reopen Properties from File menu and verify that no change occurs to PTU settings. 7) Close PTU Application.	1) The PTU main window pops up and the “There has been an error with the RS-232 serial comm port.” message pops up. 2) The message closes and the PTU Properties window pops up. 3) The Properties window is closed. 4) The PTU Properties window pops up. 5) The Properties window is closed. 6) The PTU Properties window pops up. No changes occur to PTU settings. 7) The PTU main window is closed.		
5	TC3.2.5 Communication Speed Verification	1) In the PTU Simulator, change communication speed to 38400 and clean the communication log by clicking New icon. Start the PTU. It will try to login automatically. 2) Wait 10 seconds. 3) In the PTU Simulator, change communication speed to 115200. In the PTU Main window tool bar, click the Login button. 4) Close PTU Application.	1) The PTU main window pops up. 2) No package is received in the PTU Simulator. 3) The PTU Simulator receives 3 PTU_LoginReq command to DCU (3A xx xx xx 46 44 65 6E 76 65 72 2C 20 50 54 55 20 44 6F 6F 72 20 4D 6F 64 75 6C 65 20 53 79 73 74 65 6D 2C 20 56 53 52 53 xx xx) from PTU. 4) The PTU main window is closed.		

5.3.4 TC3.3 Login/Logout

Special Procedural Requirements: Make sure all Auto check-boxes in PTU Seq. and DCU Seq. are checked, and PTU Access is administrator.

Measurements: Self-explanatory.

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
1	TC3.3.1 PTU Application logs in automatically with the DCU when launched	1) Make sure the PTU Application is closed. In the PTU Simulator, compose and enable to send it back to the PTU: 1. PTU_Login 2. PTU_Location_ID 3. PTU_SciVersion 4. PTU_Version (00 10 00 07) Start PTU Application. 2) In PTU Simulator compose: PTU_Logout and enable to send it back to the PTU. Click “Logout” button on the tool bar in PTU Application.	1) The PTU Simulator receives PTU_LoginReq command (3A xx xx xx 46 44 65 6E 76 65 72 2C 20 50 54 55 20 44 6F 6F 72 20 4D 6F 64 75 6C 65 20 53 79 73 74 65 6D 2C 20 56 53 52 53 xx xx) from PTU. The PTU Simulator sends automatically PTU_Login (01 37 xx xx xx 47 03 44 65 6E 76 65 72 2C 20 44 6F 6F 72 20 4D 6F 64 75 6C 65 20 53 79 73 74 65 6D 2C 20 56 53 52 53 xx xx 00). In the PTU status bar, the disconnected icon is changed to the connected icon ( ). The PTU sends sequentially PTU_LocationReq (30), PTU_VersionReq (0F), and PTU_VersionReq (0F). The PTU Simulator answers, respectively, PTU_Location_ID (31), PTU_SciVersion (11), and PTU_Version (10). 2) In the PTU status bar, the connected icon is changed to the disconnected icon ( ).		
2	TC3.3.2 DCU Boot Loader Sends a Valid Login	1) In the PTU File Menu, select Properties. 2) In Login Option, select Boot Loader and click OK button. 3) In the PTU Simulator, compose:	1) The PTU Properties window pops up 2) The PTU Properties window is closed. 3) The PTU Simulator receives PTU_LoginReq command (3A xx xx xx 46 44 65 6E 76 65 72		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
		1. PTU_Login in channel C1 with status_code = 2 (Login DCU Boot Loader OK) (02 44 65 6E 76 65 72 2C 20 44 6F 6F 72 20 4D 6F 64 75 6C 65 20 53 79 73 74 65 6D 2C 20 56 53 52 53) 2. PTU_SciVersion in channel C1 and enable to send them back to the PTU. In the PTU Main window status bar (at the bottom), click on the disconnected icon ( ).	2C 20 50 54 55 20 44 6F 6F 72 20 4D 6F 64 75 6C 65 20 53 79 73 74 65 6D 2C 20 56 53 52 53 xx xx) from PTU. The PTU Simulator sends automatically PTU_Login (01 37 xx xx xx 47 02 44 65 6E 76 65 72 2C 20 44 6F 6F 72 20 4D 6F 64 75 6C 65 20 53 79 73 74 65 6D 2C 20 56 53 52 53 xx xx 00). In the PTU status bar, the disconnected icon is changed to the connected icon ( ).		
3	TC3.3.3 DCU Boot Loader Valid Command Testing	In the PTU Application File menu, select “New Project…” (don’t save previous project).	The PTU Control Menu pops up. Verify the following issues: • On the PTU Control Menu, the following are **disabled**: Test, Data Monitoring Snapshot Download, Snapshot Erasure, Log Download, Browser, Graph Download System Counters Download Log Erasure • On the PTU tool bar, the following are **disabled**: Log Download, Snapshot Download, Graph Download, Log Erasure, Snapshot Erasure, Download All. • In the PTU Maintenance menu, the following are **disabled**: Test, Data Monitoring Snapshot Download, Log Download,		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
			Door Move Graph, Graph Download, Door Commands, System Counters Download, Browser Download All • In the Administration menu, the following are **disabled**: Log Erasure, DCU Configure, Snapshot Erasure, Send Flash File. • In Tools menu, the following are **disabled**: DCU Time Update EDF&OPDF.		
4	TC3.3.4 DCU Boot Loader Sends a Valid Logout	1) Continuing from the previous step, in the PTU Simulator, compose PTU_Logout in channel C1 with status_code = 02 (Logout DCU Boot Loader OK) and enable to send it back to the PTU. In the PTU Main window status bar (at the bottom), click on the connected icon ( ).	1) The PTU Simulator will receive PTU_LogoutReq command (07 xx xx xx 48 xx xx xx) from PTU. In the PTU status bar, the connected icon is changed to the disconnected icon ( ).		
5	TC3.3.5 DCU Boot Loader Sends an Invalid Login	1) In the PTU Simulator, compose PTU_Login in channel C1 with status_code = 1 (Error occurred) (01 44 65 6E 76 65 72 2C 20 44 6F 6F 72 20 4D 6F 64 75 6C 65 20 53 79 73 74 65 6D 2C 20 56 53 52 53) and enable to send it back to the PTU. In the PTU Main window status bar (at the bottom), click on the disconnected icon ( ). 2) The PTU Simulator sends automatically PTU_Login with error occurred status back to PTU (01 37 xx xx 00 47 01 44 65 6E 76 65 72 2C 20 44 6F 6F 72 20 4D 6F 64 75 6C 65 20 53 79 73 74 65 6D 2C 20 56 53 52 53 xx xx 00).	1) The PTU Simulator receives PTU_LoginReq command (3A xx xx 00 46 44 65 6E 76 65 72 2C 20 50 54 55 20 44 6F 6F 72 20 4D 6F 64 75 6C 65 20 53 79 73 74 65 6D 2C 20 56 53 52 53 xx xx) from PTU. 2) PTU pops up the message box “DCU replied with a login error.” No change in the PTU status bar.		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
		3) Click 'OK' to close message window. Close PTU Application & do not save the project.	3) The error message is closed. The PTU main window is closed.		
6	TC3.3.6 Physical Disconnection from DCU after Login to DCU Boot Loader	1) Redo the earlier test Case for 'DCU Boot Loader sends a valid Login'. Disconnect the PTU cable from the RS-232 port on the PTU computer (not the usb-to-serial adapter). 2) Close PTU Application. Reconnect the PTU cable to the PTU computer.	1) In the PTU status bar, the connected icon is changed to the disconnected icon ( ). 2) The PTU main window is closed.		
7	TC3.3.7 PTU Exit after Login to DCU Boot Loader	Redo the earlier test case for 'DCU Boot Loader sends a valid Login'. In the PTU Simulator, compose PTU_Logout in channel C1 with status_code = 02 (Logout DCU Boot Loader OK) and enable to send them back to the PTU. In the PTU File menu, select Exit and exit without saving the project.	The PTU window is closed. Verify that the following logs are displayed on PTU Simulator: 1. The PTU Simulator receives PTU_LogoutReq command (07 xx xx 00 48 xx xx). 2. The PTU Simulator sends PTU_Logout (01 07 xx xx 00 49 02 xx xx 00).		
8	TC3.3.8 DCU Application Sends a Valid Login	1) In the PTU Simulator File menu, select New. Start the PTU Application. 2) In the PTU File Menu, select Properties. 3) In Login Option, select DCU Application and click OK button. 4) In the PTU Simulator, compose: 1. PTU_Login in channel C1 with status_code = 3 (Login DCU Application OK) (03 44 65 6E 76 65 72 2C 20 44 6F 6F 72 20 4D 6F 64 75 6C 65 20 53 79 73 74 65 6D 2C 20 56 53 52 53) 2. PTU_Location_ID 3. PTU_SciVersion 4. PTU_Version (00 10 00 07) and enable to send it back to the PTU.	1) The PTU main window pops up. 2) The PTU Properties window pops up 3) The PTU Properties window is closed. 4) The PTU Simulator receives PTU_LoginReq command (3A xx xx xx 46 44 65 6E 76 65 72 2C 20 50 54 55 20 44 6F 6F 72 20 4D 6F 64 75 6C 65 20 53 79 73 74 65 6D 2C 20 56 53 52 53 xx xx) from PTU.		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
		In the PTU Main window status bar (at the bottom), click on the disconnected icon ( ). 5) The PTU Simulator sends automatically PTU_Login (01 37 xx xx 00 47 03 44 65 6E 76 65 72 2C 20 44 6F 6F 72 20 4D 6F 64 75 6C 65 20 53 79 73 74 65 6D 2C 20 56 53 52 53 xx xx 00).	5) In the PTU status bar, the disconnected icon is changed to the connected icon ( ).		
9	TC3.3.9 DCU Application Valid Command Testing	In the PTU Simulator, compose the following command and data in channel C1 and enable to send: 1. PTU_Login 2. PTU_LocationID 3. PTU_SciVersion 4. PTU_Version (00 10 00 07) In the PTU Application File menu, select New (don't save previous project).	The PTU Control Menu pops up. Verify the following issues: • On the PTU Control Menu, **nothing** is **disabled** (except Send Flash file if no flash file is opened) • On the PTU tool bar, **nothing** is **disabled**, **except** the "Stop" button. • In the PTU Maintenance menu, **nothing** is **disabled** • In the Administration menu, **nothing** is **disabled** (except Send Flash file if no flash file is opened)		
10	TC3.3.10 DCU Application Sends a Valid Logout	1) In the PTU Simulator, compose PTU_Logout in channel C1 with status_code = 03 (Logout DCU Application OK) and enable to send it back to the PTU. In the PTU Main window status bar (at the bottom), click on the connected icon ( ).	1) The PTU Simulator will receive PTU_LogoutReq command (06 xx xx 00 48 xx xx) from PTU. In the PTU status bar, the connected icon is changed to the disconnected icon ( ).		
11	TC3.3.11 DCU Application Sends an Invalid Login	1) In the PTU Simulator, compose PTU_Login in channel C1 with status_code = 1 (Error occurred) (01 44 65 6E 76 65 72 2C 20 44 6F 6F 72 20 4D 6F 64 75 6C 65 20 53 79 73 74 65 6D 2C 20 56 53 52 53) and enable to send it back to the PTU. In the PTU Main window status bar (at the bottom), click on the disconnected icon ( ). 2) The PTU Simulator sends automatically PTU_Login with error occurred status back to PTU (01 37 xx xx 00 47 01 44 65 6E 76 65 72	1) The PTU Simulator receives PTU_LoginReq command (3A xx xx 00 46 44 65 6E 76 65 72 2C 20 50 54 55 20 44 6F 6F 72 20 4D 6F 64 75 6C 65 20 53 79 73 74 65 6D 2C 20 56 53 52 53 xx xx) from PTU. 2) PTU pops up with a login error message. No change in the PTU status bar.		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
		2C 20 44 6F 6F 72 20 4D 6F 64 75 6C 65 20 53 79 73 74 65 6D 2C 20 56 53 52 53 xx xx 00). 3) Click 'OK' to close message window. Close PTU Application and do not save the project.	3) The error message is closed. The PTU main window is closed.		
12	TC3.3.12 Physical Disconnection from DCU after Login to DCU Application	1) Redo the earlier test case titled 'DCU Application sends a valid Login'. Remove the RS232 connector from the PTU computer. 2) Close the PTU Application. Connect the PTU cable to the PTU computer.	1) In the PTU status bar, the connected icons are changed to the disconnected icons ( ). 2) The PTU main window is closed.		
13	TC3.3.13 PTU Exit after Login to DCU Application	Redo the earlier test case titled 'DCU Application sends a valid Login'. In PTU Simulator, compose PTU_Logout and enable to send it in channel C1. In the PTU File menu, select Exit (don't save current project).	The PTU window is closed. The following logs are displayed on PTU Simulator: 1. The PTU Simulator receives PTU_LogoutReq command (07 xx xx 00 48 xx xx xx). 2. The PTU Simulator sends PTU_Logout (01 07 xx xx 00 49 03 xx xx 00).		
14	TC3.3.14 PTU Login Option	1) In the PTU Simulator File menu, select New. In the PTU File Menu, select Properties. 2) In Login Option, select DCU Application and click OK button. 3) In the PTU Simulator, compose the following commands in channel C1 and enable to send them back to the PTU: 1. PTU_Login with status_code = 2 (Login DCU Boot Loader OK) (02 44 65 6E 76 65 72 2C 20 44 6F 6F 72 20 4D 6F 64 75 6C 65 20 53 79 73 74 65 6D 2C 20 56 53 52 53). 2. PTU_Logout with status_code = 2 (Logout DCU Boot Loader OK) (02). In the PTU Main window status bar (at the bottom), click on the disconnected icon ( ). 4) The PTU Simulator sends automatically PTU_Login command with status_code = 2 (Login DCU Boot Loader OK) (01 37 xx xx 00	1) The PTU Properties window pops up. 2) The PTU Properties window is closed. 3) The PTU Simulator will receive PTU_LoginReq command (3A xx xx 00 46 44 65 6E 76 65 72 2C 20 50 54 55 20 44 6F 6F 72 20 4D 6F 64 75 6C 65 20 53 79 73 74 65 6D 2C 20 56 53 52 53 xx xx) from PTU. 4) The PTU Simulator will receive PTU_LogoutReq command (07 xx xx 00 48 xx xx) from PTU.		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
		47 02 44 65 6E 76 65 72 2C 20 44 6F 6F 72 20 4D 6F 64 75 6C 65 20 53 79 73 74 65 6D 2C 20 56 53 52 53 xx xx 00) 5) The PTU Simulator sends automatically PTU_Logout command with status_code = 2 (Logout DCU Boot Loader OK) (01 07 xx xx 00 49 02 xx xx 00) 6) In the PTU Simulator, compose: 1. PTU_Login in channel C1 with status_code = 3 (Login DCU Application OK) (03 44 65 6E 76 65 72 2C 20 44 6F 6F 72 20 4D 6F 64 75 6C 65 20 53 79 73 74 65 6D 2C 20 56 53 52 53) 2. PTU_SciVersion 3. PTU_Location_ID 4. PTU_Version (00 10 00 07) and enable to send it back to the PTU. In the PTU Main window tool bar, click login icon. 7) The PTU Simulator sends automatically PTU_Login command with status_code = 3 (Login DCU Application OK) (01 37 xx xx 00 47 03 44 65 6E 76 65 72 2C 20 44 6F 6F 72 20 4D 6F 64 75 6C 65 20 53 79 73 74 65 6D 2C 20 56 53 52 53 xx xx 00). 8) Close PTU Application.	5) In status bar, the PTU shows the PTU logout icon ( ). 6) The PTU Simulator will receive PTU_LoginReq command (3A xx xx 00 46 44 65 6E 76 65 72 2C 20 50 54 55 20 44 6F 6F 72 20 4D 6F 64 75 6C 65 20 53 79 73 74 65 6D 2C 20 56 53 52 53 xx xx) from PTU. 7) In status bar, the PTU shows the PTU logged into the DCU application ( ). 8) The PTU main window is closed.		
15	TC3.3.15 PTU Sends Logout and Reboot command	Tested along with test case 1 “Application Upgrade Request” in Section 5.3.25.	See output of test case 1 of Section 5.3.25.		

5.3.5 TC3.4 Communication Error Process

Special Procedural Requirements: None.
Measurements: Self-explanatory.

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
1	TC3.4.1 DCU to PTU CRC Error	1) In the PTU Simulator, click TX label to disable CRC. 2) Start the PTU Application. 3) In the PTU Simulator, compose PTU_Login in channel C1 with status_code = 3 (Login DCU Application OK) (03 44 65 6E 76 65 72 2C 20 44 6F 6F 72 20 4D 6F 64 75 6C 65 20 53 79 73 74 65 6D 2C 20 56 53 52 53) and enable to send it back to the PTU. In the PTU Main window status bar (at the bottom), click on the disconnected icon ( ). 4) The PTU Simulator sends automatically PTU_Login (01 37 xx xx 00 47 03 44 65 6E 76 65 72 2C 20 44 6F 6F 72 20 4D 6F 64 75 6C 65 20 53 79 73 74 65 6D 2C 20 56 53 52 53 00 00 00). 5) Verify that the time delay between PTU_login from the PTU Simulator (previous step) and the first PTU_LoginReq from the PTU Application (previous step) is more than 200ms and less than 2000ms. 6) Close PTU Application. In the PTU Simulator, click TX label to enable CRC for a later test.	1) The PTU Simulator TX text becomes red. 2) The PTU main window pops up. 3) The PTU Simulator will receive PTU_LoginReq command (3A xx xx 00 46 44 65 6E 76 65 72 2C 20 50 54 55 20 44 6F 6F 72 20 4D 6F 64 75 6C 65 20 53 79 73 74 65 6D 2C 20 56 53 52 53 xx xx). 4) The PTU Simulator will receive PTU_LoginReq command (3A xx xx 00 46 44 65 6E 76 65 72 2C 20 50 54 55 20 44 6F 6F 72 20 4D 6F 64 75 6C 65 20 53 79 73 74 65 6D 2C 20 56 53 52 53 xx xx). After 500ms the PTU Simulator will receive PTU_LoginReq command again (3A xx xx 00 46 44 65 6E 76 65 72 2C 20 50 54 55 20 44 6F 6F 72 20 4D 6F 64 75 6C 65 20 53 79 73 74 65 6D 2C 20 56 53 52 53 xx xx). 5) The time delay between packets is between 200ms and 2000ms. 6) The PTU main window is closed. The PTU Simulator TX text becomes black.		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
2	TC3.4.2 Communication Time-Out Triggering Retries	1) Redo Test Case #8 of section 5.3.4 [DCU Application sends a valid Login]. In PTU Tools menu, select DCU Time. 2) Use the PTU Simulator time stamps to verify message delays. 3) Close PTU Application.	1) The DCU Time window pops up. The PTU Simulator receives 3 times PTU_TimeReq command (06 xx xx 00 40 xx xx). The PTU automatically logs out and shows the disconnect icon. 2) The time delay between each PTU_TimeReq packet is between 200ms and 2000ms. 3) The PTU main window is closed.		
3	TC3.4.3 Communication Error after 3 Tries	See Test Case #2 [Communication time-out triggering retries].			

5.3.6 TC3.5 EDF and OPDF

Special Procedural Requirements: None.
Measurements: Self-explanatory.

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
1	TC3.5.1 EDF Update and Browse	1) In the PTU Simulator, open the PTU simulation file **EDF.bin** for sending (it is located in the same directory of PTU Simulator software). In the PTU Simulator, compose the following 4 commands and data in channel C1, and enable to send: 1. PTU_Login (03 44 65 6E 76 65 72 2C 20 44 6F 6F 72 20 4D 6F 64 75 6C 65 20 53 79 73 74 65 6D 2C 20 56 53 52 53) 2. PTU_LocationID 3. PTU_SciVersion 4. PTU_Version (7F FF 00 00). In the PTU Main window status bar (at the bottom), click on the disconnected icon ( ). 2) Close PTU Application and do not save the project.	1) The PTU starts to download the EDF from the DCU, and a progress bar displays the status of the download process. When the downloading is finished, the EDF window pops up. Browse and verify EDF with the Table A-3. 2) The PTU main window is closed.		
2	TC3.5.2 EDF Download	Verify if the PTU Simulator communication log for Test Case #1 [EDF Update and Browse] is compliant with Appendix A, Figure A-13.	1. The following operations are found in the log: 2. The PTU sends PTU_FileReq with file index =2 and request=FR_OPEN_TO_SEND (0) (12 xx xx 00 03 00 00 02 00 00 00 00 00 00 00 00 00 xx xx). 3. The PTU Simulator sends PTU_FileStat with file status FS_XFER_UNDERWAY(value 4) back to the PTU (01 22 xx xx 00 04 04 00 00 00 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 xx xx xx xx 00 00 xx xx 00). 4. The PTU sends PTU_NextPackReq to the PTU Simulator (06 xx xx 00 20 xx xx). 5. The PTU Simulator sends PTU_NextPack with file data to the PTU.		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
			6. The step 3 and 4 are repeated until the file is finished. 7. After the PTU receives PTU_LastPack, it sends PTU_FileReq with file index =2 and request=FR_CLOSE_FILE(2) (12 xx xx 00 03 02 00 02 00 00 00 00 00 00 00 00 00 xx xx). 8. The PTU Simulator sends PTU_FileStat with FS_XFER_OK (0) to the PTU (01 22 xx xx 00 04 00 00 00 00 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 xx xx xx xx 00 00 xx xx 00).		
3	TC3.5.3 OPDF Update and Browse	1) In the PTU Simulator, open the PTU simulation file OPDF.bin for sending (it is located in the same directory of PTU_Simulator software). In the PTU Simulator, compose the following commands and data in channel C1, and enable to send: 1. PTU_Login (03 44 65 6E 76 65 72 2C 20 44 6F 6F 72 20 4D 6F 64 75 6C 65 20 53 79 73 74 65 6D 2C 20 56 53 52 53) 2. PTU_Location_ID 3. PTU_SciVersion 4. PTU_Version (00 00 7F FF). In the PTU Main window status bar (at the bottom), click on the disconnected icon ( ). 2) Close PTU Application & do not save the project.	1) The PTU starts to download the EDF then the OPDF from the DCU, and a progress bar displays the status of the download process. After the download finishes, the OPDF window pops up. Browse and verify OPDF with the Table A-4. 2) The PTU main window is closed.		
4	TC3.5.4 OPDF Download	Verify if the PTU Simulator communication log for Test Case #3 [OPDF Update and Browse] compliant with Appendix A, Figure A-13.	The following operations are found in the log: 1. The PTU sends PTU_FileReq with file index =4 and request=FR_OPEN_TO_SEND (0) (12 xx xx 00 03 00 00 04 00 00 00 00 00 00 00 00 00 xx xx). 2. The PTU Simulator sends PTU_FileStat with file status FS_XFER_UNDERWAY(value 4) back to the PTU (01 22 xx xx 00 04 04 00 00 00 00 04 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 xx xx xx xx 00 00 xx xx 00).		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
			3. The PTU sends PTU_NextPackReq to the PTU Simulator (06 xx xx 00 20 xx xx). 4. The PTU Simulator sends PTU_NextPack with file data to the PTU. 5. The step 3 and 4 are repeated until the file is finished. 6. After the PTU receives PTU_LastPack, it sends PTU_FileReq with file index =4 and request=FR_CLOSE_FILE(2) (12 xx xx 00 03 02 00 04 00 00 00 00 00 00 00 00 00 xx xx). 7. The PTU Simulator sends PTU_FileStat with FS_XFER_OK (0) to the PTU (01 22 xx xx 00 04 00 00 00 00 00 04 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 xx xx xx xx 00 00 xx xx 00).		

5.3.7 TC3.6 Request DCU Code Status

Special Procedural Requirements: None.
Measurements: Self-explanatory.

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
1	TC3.6.1 Request the Code Status of the DCU Boot Loader	1) Do Test Case #8 of section 5.3.4 [DCU Application sends a valid Login]. In the PTU Simulator, compose the following commands and data in channel C1, and enable to send: 1. PTU_DCUCodeStatus (01 00 00 80 00 12 34 56 78) 2. PTU_DCUCodeStatus (02 00 01 00 00 9A BC DE F0) 3. PTU_DCUCodeStatus (03 00 00 76 B7 AE 12 87 65) From PTU Tools menu, select DCU Code Status. 2) Close PTU Application.	1) The DCU Code Status pops up. The follow software code information is shown on the Code Status window: 1. DCU Boot Loader, size=32768 and CRC=0x12345678. 2. DCU Application, size=65536 and CRC=0x9ABCDEF0. 3. Neuron Application, size=30391 and CRC=0xAE128765. 2) The PTU main window is closed.		
2	TC3.6.2 Request the Code Status of the DCU Application	See Test Case #1 [Request the code status of the DCU Boot Loader].			
3	TC3.6.3 Request the Code Status of the DCU Neuron Application	See Test Case #1 [Request the code status of the DCU Boot Loader].			

5.3.8 TC3.7 Read Location Identification and Car Number

Special Procedural Requirements: None.
Measurements: Self-explanatory.

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
1	TC3.7.1 Location ID Request	1) Redo steps 1 to 5 of test case #8 of section 5.3.4 [DCU Application sends a valid Login] except in the PTU Simulator, compose the following 3 commands in channel C1: 1. PTU_SciVersion, 2. PTU_LocationID (01 04 D2), 3. PTU_Version (00 10 00 07) And enable to send. In the PTU File menu, select New (don't save previous project). 2) Change PTU_LocationID to the following different location and re-do previous test step. Verify the location on the control menu. PTU_LocationID (02 00 64) PTU_LocationID (03 00 C8) PTU_LocationID (04 01 2C)	1) The PTU is connected to the DCU. The PTU Control Menu pops up. The DCU location is shown on the control menu: DCU 01, Car No. 1234. 2) The DCU location is shown on the left side control menu: DCU 02, Car No. 100. DCU 03, Car No. 200. DCU 04, Car No. 300.		
2	TC3.7.2 Location ID Received	See Test Case #1 [Location ID request].			
3	TC3.7.3 Location ID Displayed on the left side control menu when PTU Application starts	See Test Case #1 [Location ID request].			
4	TC3.7.4 Car Number displayed on the left side control menu when PTU Application starts	See Test Case #1 [Location ID request].			

5.3.9 TC3.8 Software Versions

Special Procedural Requirements: None.
Measurements: Self-explanatory.

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
1	TC3.8.1 Versions Request from PTU	In the PTU Simulator, compose the following commands and data and enable to send at channel C1: 1. PTU_Login 2. PTU_LocationID (01 04 D2) (DCU 01, Car No. 1234) 3. PTU_SciVersion 4. PTU_Version (00 10 00 07) In the PTU, click Login button.	The PTU Control Menu pops up with following information: DCU Location – DCU 01, Car No. 1234 The System SW Version window popups with the following information: DCU SCI version list (for example): APPLICATION, 1.00, 03-19-2008 BOOT LOADER, 1.00, 02-15-2008 NEURON APP, 1.00, 03-19-2008 CPLD, 1.00,		
2	TC3.8.2 Versions Retrieved by PTU	See Test Case #1 [Versions request from PTU].			
3	TC3.8.3 Display the Following Versions: DCU Application, DCU Boot Loader, DCU Neuron Application, and CPLD.	See Test Case #1 [Versions request from PTU]			

5.3.10 TC3.9 Real-Time Monitoring

Special Procedural Requirements: None.
Measurements: Self-explanatory.

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
1	TC3.9.1 Start Real-Time Data Monitoring Request Note : The next TC should be executed immediately after this one	Do Test Case #1 of section 5.3.9 [Versions request from PTU]. In PTU Maintenance→Other Functions menu, select Browser	The DCU Browser window pops up and all Operating Parameters are listed in the Browser table (See in Appendix A, Figure A-7). The PTU Simulator continuously receives Data Requests from the PTU.		
2	TC3.9.2 Real-Time Data Monitoring Displayed	1) In the PTU Simulator, compose the following commands and data at channel C3 (right-click twice on channel button): 1. PTU_Data (FF FF FF FF FF FF FF FF FF FF FF FF FF FF) 2. PTU_Data (00 00 00 00 00 00 00 00 00 00 00 00 00 00). Click on no. 1 command channel button. 2) Click on no. 2 command channel button.	1) The Parameters value will be displayed in the PTU Browser table (See Appendix A, Figure A-8). All the binary values = 1, motor current = 25.4, Door Panel Opening Position =127, DCU Temperature = 195, Motor Drive Voltage = 255. All binary values are “1” 2) The Parameters value will be displayed in the PTU Browser table. The Motor Current = –25.6 A, Motor Drive Voltage = 0, DCU Temperature = –60, Door Opening Position = –128 All binary values are “0”		
3	TC3.9.3 Stop Real-Time Data Monitoring Request	1) Close PTU Browser window. 2) Close PTU Application and do not save the project.	1) The PTU Application stops to send polling command to the PTU Simulator and the PTU Browser window is closed. 2) The PTU main window is closed.		

5.3.11 TC3.10 Retrieve Self Test

Special Procedural Requirements: None.
Measurements: Self-explanatory.

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
1	TC3.10.1 Self Test Results Request	1) Do Test Case #1 of section 5.3.9 [Versions request from PTU]. In the PTU Simulator, compose the following commands and data and enable to send at channel C1: PTU_SelftestStatus (FF FF) (All self-tests pass) Click “TEST” button on PTU Control Menu. 2) The PTU Simulator sends automatically PTU_SelftestStatus (FF FF) to PTU. 3) In the PTU Simulator, compose the following commands and data and enable to send at channel C1: PTU_SelftestStatus (00 00) (All self-tests fail) Click “Redo” button on PTU Self Test window. 4) The PTU Simulator sends automatically PTU_SelftestStatus (00 00) to PTU. 5) Close PTU Application and do not save the project.	1) The Self Test window pops up. The PTU Simulator receives DCU PTU_SelftestReq command (xx xx xx 00 42 xx xx). 2) The Self Test window shows all self tests “pass”. 3) The PTU Simulator receives DCU PTU_SelftestReq command (xx xx xx 00 42 xx xx). 4) The Self Test window shows all self tests “fail”. 5) The PTU main window is closed.		
2	TC3.10.2 Self Test Results Display	See Test Case #1 [Self Test result request].			

5.3.12 TC3.11 DCU Status

Special Procedural Requirements: None.
Measurements: Self-explanatory.

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
1	TC3.11.1 DCU Status Request	Execute Test Case #1 of section 5.3.9 [Versions request from PTU]. On the PTU Control Menu, click “Data Monitoring” button.	The DCU Status window pops up. The PTU Simulator receives PTU_DataReq (06 xx 00 00 0D xx xx) command from PTU Application every 200 milliseconds.		
2	TC3.11.2 DCU Status Information Displayed	While the Status window is still open in PTU, in the PTU Simulator, compose the following commands and data at channel C3 (right-click on C1 button to change channel). 1) PTU_Data (00 00 00 00 00 00 00 00 00 00 00 00 00). Click on no. 1 command channel button. 2) PTU_Data (FF FF FF FF FF FF FF FF FF FF FF FF FF). Click on no. 2 command channel button.	1) All indicators are turned off, Motor Current is set to –25.6 A, Motor Drive Voltage shall be 0, DCU Temperature –60, and Door Opening Position –128. (See Appendix A, Figure A-9). 2) All indicators will turn on; Motor Drive Voltage (255), Motor Current (A) (25.4), DCU Temperature (195), and Door Position (127%).		
3	TC3.11.3 Event Synchronization	1) In the PTU Simulator, compose the following commands and data at channel C2: 1. PTU_Event (00 32 00 00 C1 44 4D 43 52 31 32) 2. PTU_Event (00 32 00 00 80 44 4D 43 52 31 32) Click on no. 1 command channel button. 2) Click on no. 2 command channel button.	1) The event 50 (Set) is added into event table. 2) The event 50 (Reset) is added into event table.		
4	TC3.11.4 System status and functional information shall be provided in a graphical format, superimposed over a system schematic or diagrammatic. When a graphic presentation is not practical,	Verify that all status elements are either graphical and/or provided in English language with relevant engineering units.	**Element** Running Warning Failure Door Summary Relay High Side Door Summary Relay Low Side		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
	system status and functional information shall be provided in English language statements and relevant engineering units, and shall be formatted for simple readability.		5V Logic Supply 5V Encoder Supply 15V Motor Supply V-Gate Open Supply V-Gate Close Supply Motor Drive Voltage Door Closing Visual Alarm Door Closing Audible Alarm Passenger Door Enable Light CPU Open Interlock Solenoid Lock Spare Output Door Open (+) Train Line Door Open (-) Train Line Door Close Train Line Zero Speed (+) Car Line Zero Speed (-) Car Line Passenger Door Open Pushbutton Passenger Door Enable (+) Train Line Passenger Door Enable (-) Train Line Emergency Handle Sensing Switch Door Cut-out Mechanical Lock Normal Panel Left Open Panel Right Open Solenoid Lock 1 Unlocked Solenoid Lock 2 Unlocked CPLD Open Interlock Powered CPLD Solenoid Interlock Powered Overcurrent Shutdown Spare 1 Input Spare 2 Input Motor Current DCU Temperature Door Opening Position Crew Switch Open Crew Switch Enable Crew Switch Close		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
5	TC3.11.5 System status and functional information conform to the following principles: - In the case of units of temperature, absolute units (Kelvin, Rankin) shall not be used, and prefixing the For the ºC symbol with a small circle (degree symbol) shall not be used: instead, "deg" or "degrees" shall be the prefix, e.g. "deg F", "degrees C" etc	1) Make sure that the status window respects the following rules: - In the case of units of temperature, absolute units (Kelvin, Rankin) shall not be used, and prefixing the For C symbol with a small circle (degree symbol) shall not be used: instead, "deg" or "degrees" shall be the prefix, e.g. "deg F", "degrees C" etc 2) Close DCU Status window. Close PTU Application and do not save the project.	1) The Status window respects the rules. 2) The DCU Status window is closed. The PTU main window is closed.		

5.3.13 TC3.12 Door Open Time Configuration

Special Procedural Requirements: Make sure the PTU Access is in administrator.
Measurements: Self-explanatory.

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
1	TC3.12.1 Get Door Open Time Configuration from DCU TC3.12.2 Validate Door Open Time Configuration Entered by User TC3.12.3 Send Door Open Time Configuration to DCU TC3.12.4 Door Open Time Configuration Reception Confirmation from DCU	1) Do Test Case #8 of section 5.3.4 [DCU Application sends a valid Login]. In the PTU Simulator, compose the following commands and data and enable to send at channel C1: 1. PTU_DCUConfigStatus (01 01 00 14) (Open Time 2.0 seconds) 2. PTU_DCUConfigStatus (01 02 00 19) (Close Time 2.5 seconds) 3. PTU_DCUConfigStatus (01 03 00 02) (Obstruction Retries Times 2) 4. PTU_DCUConfigStatus (01 04 00 14) (Obstruction Re-Open Disance 20 inches). 5. PTU_DCUConfigStatus (01 05 00 03) (Delay before reclosing 3 seconds). 6. PTU_DCUConfigStatus (01 06 00 04) (Delay before door closing 4 seconds). 7. PTU_DCUConfigStatus (01 07 00 07) (Door Open Duration for Passenger Door Enable 7 minutes). From Administration -> Other Functions menu, select DCU Configure. 2) Click on the configuration table, a combo box pops up and it allows the PTU user to change the current configuration. Verify the valid data for each configuration item in combo box.	1) The DCU Configuration window pops up with configuration table. PTU receives DCU configuration information from PTU Simulator and displays on the configuration table: Opening Time (s): **2.0** Closing Time (s): **2.5** Obstruction Retries Times: **2** Obstruction Re-Open Distance ("): **20** Delay before reclosing (s): **3** Delay before door closing (s): **4** Door Open Duration for Passenger Door Enable (min): **7** 2) Door Opening Time (s): 2.0, 2.5, 3.0 Door Closing Time (s): 2.0, 2.5, 3.0 Obstruction Retries: 0, 1, 2, 3, 4, 5 Obstruction Re-Open Disance (“): 0, 3, 5, 10, 15, 20, 25 Delay before reclosing (s): 0 to 5, in steps of 1 Delay before door closing (s): 0 to 5, in steps of 1 Door Open Duration for Passenger Door Enable (min): 0 to 10, in steps of 1		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
		3) In the PTU Simulator, compose the following commands and data and enable to send at channel C1: 1. PTU_DCUConfigStatus (01 01 00 19) (Open Time 2.5 seconds) 2. PTU_DCUConfigStatus (01 02 00 1E) (Close Time 3.0 seconds) 3. PTU_DCUConfigStatus (01 03 00 04) (Obstruction Retries Times 4) 4. PTU_DCUConfigStatus (01 04 00 03) (Obstruction Re-Open Disance 3 inches). 5. PTU_DCUConfigStatus (01 05 00 01) (Delay before reclosing 1 seconds). 6. PTU_DCUConfigStatus (01 06 00 03) (Delay before door closing 3 seconds). 7. PTU_DCUConfigStatus (01 07 00 05) (Door Open Duration for Passenger Door Enable 5 minutes). In the PTU, set the above configuration values and click Apply.	3) The following communication logs are found in the PTU Simulator: [RX]:0A xx xx 00 4C 02 01 00 19 xx xx [TX]:01 0A xx xx 00 4D 01 01 00 19 xx xx 00 [RX]:0A xx xx 00 4C 02 02 00 1E xx xx [TX]:01 0A xx xx 00 4D 01 02 00 1E xx xx 00 [RX]:0A xx xx 00 4C 02 03 00 04 xx xx [TX]:01 0A xx xx 00 4D 01 03 00 04 xx xx 00 [RX]:0A xx xx 01 4C 02 04 00 03 xx xx [TX]:01 0A xx xx 01 4D 01 04 00 03 xx xx 00 [RX]:0A xx xx 01 4C 02 05 00 01 xx xx [TX]:01 0A xx xx 01 4D 01 05 00 01 xx xx 00 [RX]:0A xx xx 01 4C 02 06 00 03 xx xx [TX]:01 0A xx xx 01 4D 01 06 00 03 xx xx 00 [RX]:0A xx xx 01 4C 02 07 00 05 xx xx [TX]:01 0A xx xx 01 4D 01 07 00 05 xx xx 00		
		4) Click the Cancel button to leave DCU Configure window.	4) The DCU Configure window is closed. Message box asking to defaulting all parameters appears.		
		5) Click No	5) Message box closes and no command is sent out.		
		6) Compose the same commands as step 3 and select DCU Configure from Administration → Other Functions menu.	6) DCU Configuration window popups with the same values as step 3.		
		7) Click the Cancel button.	7) The DCU Configure window is closed. Message box asking to defaulting all parameters appears.		
		8) Compose the same commands as step 1 and click Yes.	8) PTU Simulator receives commands from PTU Application.		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
		9) Close PTU Application.	9) The PTU main window is closed.		

5.3.14 TC3.13 Door Close Time Configuration

Special Procedural Requirements: Make sure the PTU Access is in administrator.
Measurements: Self-explanatory.

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
1	TC3.13.1 Get Door Close Time Configuration from DCU TC3.13.2 Validate Door Close Time Configuration Entered by User TC3.13.3 Send Door Close Time Configuration to DCU TC3.13.4 Door Close Time Configuration Reception Confirmation from DCU	See Test Case #1 of section 5.3.13 [Get Door Open Time configuration].			

5.3.15 TC3.14 Obstruction Retries Times Configuration

Special Procedural Requirements: Make sure the PTU Access is in administrator.
Measurements: Self-explanatory.

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
1	TC3.14.1 Get Obstruction Retries Times Configuration from DCU TC3.14.2 Validate Obstruction Retries Times Configuration TC3.14.3 Send Obstruction Retries Times Configuration to DCU TC3.14.4 Obstruction Retries Times Configuration Confirmation from DCU	See Test Case #1 of section 5.3.13 [Get Door Open Time configuration].			

5.3.16 TC3.15 Obstruction Re-Open Distance Configuration

Special Procedural Requirements: Make sure the PTU Access is in administrator.
Measurements: Self-explanatory.

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
1	TC3.15.1 Get Obstruction Re-Open Distance Configuration from DCU TC3.15.2 Validate Obstruction Re-Open Distance Configuration TC3.15.3 Send Obstruction Re-Open Distance Configuration to DCU TC3.15.4 Obstruction Re-Open Distance Configuration Confirmation from DCU	See Test Case #1 of section 5.3.13 [Get Door Open Time configuration].			

5.3.17 TC3.16 Delay Before Re-closing Configuration

Special Procedural Requirements: Make sure the PTU Access is in administrator.
Measurements: Self-explanatory.

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
1	TC3.16.1 Get Delay before Re-closing Configuration from DCU TC3.16.2 Validate Delay before Re-closing Configuration TC3.16.3 Send Delay before Re-closing Configuration to DCU TC3.16.4 Delay before Re-closing	See Test Case #1 of section 5.3.13 [Get Door Open Time configuration].			

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
	Configuration Confirmation from DCU				

5.3.18 TC3.17 Delay Before Door Closing Configuration

Special Procedural Requirements: Make sure the PTU Access is in administrator.
Measurements: Self-explanatory.

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
1	TC3.17.1 Get Delay Before Door Closing Configuration from DCU TC3.17.2 Validate Delay Before Door Closing Configuration TC3.17.3 Send Delay Before Door Closing Configuration to DCU TC3.16.5 Delay Before Door Closing Configuration Confirmation from DCU	See Test Case #1 of section 5.3.13 [Get Door Open Time configuration].			

5.3.19 TC3.18 Door Open Duration for Passenger Door Enable Configuration

Special Procedural Requirements: Make sure the PTU Access is in administrator.
Measurements: Self-explanatory.

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
1	TC3.18.1 Get Door Open Duration for Passenger Door Enable Configuration from DCU TC3.18.2 Validate Door Open Duration for Passenger Door Enable Configuration TC3.18.3 Send Door Open Duration for Passenger Door Enable Configuration to DCU TC3.18.4 Door Open Duration for Passenger Door Enable Configuration Confirmation from DCU	See Test Case #1 of section 5.3.13 [Get Door Open Time configuration].			

5.3.20 TC3.19 Time of the DCU

Special Procedural Requirements: Make sure the PTU Access is in administrator.
Measurements: Self-explanatory.

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
1	TC3.19.1 DCU Time Request	1) Do Test Case #8 of section 5.3.4 [DCU Application sends a valid Login]. In the PTU Simulator, compose the following commands and data then enable to send at channel C1: 1. PTU_TimeStatus with time value 01-01-1970 00:00:00 (**00 00 00 00**) From PTU Tools menu, select DCU Time. 2) In the PTU Simulator, compose the following commands and data then enable to send at channel C1: 1. PTU_TimeStatus with time value 01-01-2000 00:00:00 (**38 6D 43 80**) Click Retrieve DCU Time button on the DCU Time window. 3) In the PTU Simulator, compose the following commands and data then enable to send at channel C1: 1.PTU_TimeStatus with time value 01-19-2038 03:14:08 (**80 00 00 00**) Click Retrieve button on the DCU Time window. 4) In the PTU Simulator, compose the following command & data, and enable to send at channel C1: 1.PTU_TimeStatus with time value 12-31-2050, 23:59:49 (**98 5B A9 75**) Click Retrieve button on the DCU Time window in the PTU Application.	1) The DCU Time window pops up. PTU displayed in the DCU Time window in format “mm-dd-yyyy hh:mm:ss”. Verify the displayed time is 01-01-1970 00:00:00. 2) Verify the time value (01-01-2000 00:00:00). 3) Verify the time value (01-19-2038 03:14:08). 4) Verify the time value (12-31-2050 23:59:49).		
2	TC3.19.2 Display Time	See Test Case #1 [DCU time request].			

5.3.21 TC3.20 Download Memory Content

Special Procedural Requirements: None.
Measurements: Self-explanatory.

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
1	TC3.20.1 Download SRAM Content Request	1) Do steps 1 to 5 of Test Case #8 of section 5.3.4 [DCU Application sends a valid Login]. In the PTU Simulator, open the PTU simulation file mem0.tmp for sending (it is located in the same directory as the PTU Simulator software). Minimize PTU Simulator window. Click Memory Dump button in the PTU Control Menu under Maintenance, Other Functions.	1) The DCU Memory Dump window pops up.		
		2) Click Memory Dump on the Tool Bar.	2) The PTU Simulator starts to download DCU SRAM contents from the DCU, and a progress bar shows the status of the download process. After the file transfer is completed, the DCU Memory values are displayed in the DCU Memory Dump window. The display format is: • Address---16 bytes data in Hex—16 bytes data in ASCII.		
		3) Verify the following operation on the Memory Dump window: • Input offset address F000. • Move the display scroll bar.	3) The display window will automatically jump to related offset address.		
		4) Close DCU Memory Dump window. Close PTU Application and do not save the project.	4) The DCU Memory Dump window is closed. The PTU main window is closed.		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
2	TC3.20.2 SRAM Download	Verify if the PTU Simulator communication log for Test Case #1 [Download Static RAM Content Request] compliant with Appendix A, Figure A-13.	The following operations are found in the log: 1. The PTU sends PTU_FileReq with file index =1 and request=FR_OPEN_TO_SEND (0) (12 xx xx xx 03 00 00 01 00 00 00 00 00 00 00 00 00 xx xx). 2. The PTU Simulator sends PTU_FileStat with file status FS_XFER_UNDERWAY(value 4) back to the PTU (01 22 xx xx xx 04 04 00 00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 90 00 00 00 xx xx 00). 3. The PTU sends PTU_NextPackReq to the PTU Simulator (06 xx xx xx 20 xx xx). 4. The PTU Simulator sends PTU_NextPack with file data to the PTU. 5. The step 3 and 4 are repeated until the file is finished. 6. After the PTU receives PTU_LastPack, it sends PTU_FileReq with file index =1 and request=FR_CLOSE_FILE(2) (12 xx xx xx 03 02 00 01 00 00 00 00 00 00 00 00 00 xx xx). 7. The PTU Simulator sends PTU_FileStat with FS_XFER_OK (0) to the PTU (01 22 xx xx xx 04 00 00 00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 90 00 00 00 xx xx 00).		
3	TC3.20.3 Validate start and end address of the memory zone to download	1) Enter 10000 in the Start Address field and 10050 in the Length field. Click Dump. 2) Type 20 in the Offset field. 3) Type 200 in the Offset field. 4) Erase all text in the Start Address field and the	1) After the memory dump is downloaded, five rows will be displayed starting at location 010000. 2) The scroll bar moves so that the top displayed row starts at location 10020 and only 3 rows are now displayed. 3) Since this offset is beyond the currently valid range, the scroll bar moves so that the top displayed row starts at location 10040. This is the only row displayed. 4) A Message window is displayed indicating that		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
		End Address field. Click Dump.	a start address must be supplied and that the request will be sent with the value 0xE000. E000 appears in the start address field.		
		5) Click OK.	5) A second Message window is displayed indicating that a valid end address must be supplied and that the request will be sent with the value 0x127FFF. 127FFF apears in the end address.		
		6) Click OK.	6) The memory dump is downloaded again with the first row starting at 00E000.		
		7) Enter 1 in the Start Address field and 0 in the Length field. Click Dump.	7) A Message window is displayed indicating that a start address must be between 0xE000 and 0x127FFF that the request will be sent with the value 0xE000. E000 appears in the start address field.		
		8) Click OK.	8) A second Message window is displayed indicating that a valid end address must be between start address and 0x127FFF and that the request will be sent with the value 0x127FFF. 127FFF appears in the end address.		
		9) Click OK.	9) The memory dump is downloaded again with the first row starting at 00E000.		
		10) Repeat steps 7 through 9 but with the value FFFFFFFF for both fields.	10) The same results as steps 7 through 9 above.		
		11) In the Start Address, End Address fields, try to enter characters that are not valid hexadecimal digits (i.e. characters other than ‘1’ to ‘9’, ‘a’ to ‘f’ or ‘A’ to ‘F’).	11) The invalid characters will not be entered into the fields.		
		12) In the end address field, try to enter a number greater than 127FFF.	12) The field will not allow the last digit to be entered and a beep will be heard.		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
		13) Close Memory Dump window. Close PTU Application and do not save the project.	13) The Memory Dump window is closed. The PTU Application main window is closed.		

5.3.22 TC3.21 Door Movement Information Graph

Special Procedural Requirements: None.
Measurements: Self-explanatory.

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
1	TC3.21.1 Download Door Graph Request	1) Do Test Case #1 of section 5.3.9 [Versions request from PTU]. In the PTU Simulator, open the PTU simulation files graph.tmp for sending (it is located in the same directory as the PTU Simulator software). Click MAINTENANCE → OTHER FUNCTIONS → GRAPH VIEW button in the PTU Control Menu. 2) Click Graph Download button on the Tool Bar. Compare the graph with Figure A-10.	1) The Door Graph window pops up. 2) The PTU Simulator starts file transferring, with a progress bar displaying the status of the transfer. After the file transfer is completed, the door graphs are displayed in the Door Graph window. The top graph contains five curves: Theoretical Speed, Speed, Position, PWM, and Current. The bottom graph contains four curves: Error, Error Slope, Error Max, and Error Min.		
2	TC3.21.2 Door Graph Download	Verify if the PTU Simulator communication log for Test Case #1 [Download Door Graph Request] is compliant with Appendix A, Figure A-13.	The following operations are found in the log: 1. The PTU sends PTU_FileReq with file index =26 and request=FR_OPEN_TO_SEND (0) (12 xx xx xx 03 00 00 1A 00 00 00 00 00 00 00 00 00 xx xx). 2. The PTU Simulator sends PTU_FileStat with file status FS_XFER_UNDERWAY(value 4) back to the PTU (01 22 xx xx xx 04 04 00 00 00 00 1A 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 xx xx xx xx 00 00 xx xx 00). 3. The PTU sends PTU_NextPackReq to the PTU Simulator (06 xx xx xx 20 xx xx).		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
			4. The PTU Simulator sends PTU_NextPack with file data to the PTU. 5. The step 3 and 4 are repeated until the file is finished. 6. After the PTU receives PTU_LastPack, it sends PTU_FileReq with file index =26 and request=FR_CLOSE_FILE(2) (12 xx xx xx 03 02 00 1A 00 00 00 00 00 00 00 00 00 xx xx). 7. The PTU Simulator sends PTU_FileStat with FS_XFER_OK (0) to the PTU (01 22 xx xx xx 04 00 00 00 00 00 1A 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 xx xx xx xx 00 00 xx xx 00).		
3	TC3.21.3 Door Graph Display	Click on the Open and Close tab Door Graph windows, verify the graph is displayed correctly.	Both graphs are identical because the same file is transferred twice. In all cases, the top graph contains five curves: Theoretical Speed, Speed, Position, PWM, and Current; and the bottom graph contains four curves, Error, Error Slope, Error Max, Error Min, Opening/Closing Time, and Configured Time.		
4	TC3.21.4 Door Graph Download Communication Error	1) On the PTU toolbar, click Graph Download button. 2) During the download, in the PTU Simulator, click the TX label to disable CRC. 3) Wait until an error message appears. 4) Close PTU Application and do not save the project. In the PTU Simulator, click TX label to enable CRC for a later test.	1) The PTU starts to download the graph file. 2) The PTU Simulator TX text color becomes red. 3) Message bubble appears because the CRC is disabled. The error message disappears. 4) The PTU main window is closed. The PTU Simulator TX text color becomes black.		

5.3.23 TC3.22 Log File and Snapshots

Special Procedural Requirements: Make sure the PTU Access is in administrator.
Measurements: Self-explanatory.

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
1	TC3.22.1 Log File Download Request	In the PTU Simulator, click New button to remove all communication logs. In the PTU Simulator, open the DCU event log simulation file **log0.tmp** for sending (it is located in the same directory as the PTU Simulator software). Do Test Case #1 of section 5.3.9 [Versions request from PTU]. Select **Download Event Log** from the Maintenance → Other Functions menu.	The PTU starts the log file download from the DCU and a progress bar is displayed. When the Log Download is completed, the event logs are displayed in the PTU Event Log table. Verify that the following steps appeared in the PTU Simulator communication logs: 1. The PTU Simulator receives PTU_FileReq command with fileindex =3 and request = FR_DIRECTORY_LOOKUP (4) (12 xx xx xx 03 04 00 03 00 00 00 00 00 00 00 00 00 xx xx). 2. The PTU Simulator sends PTU_FileStat with filesize and select_file back to the PTU (01 22 xx xx xx 04 01 00 00 00 00 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 xx xx xx xx 00 00 xx xx 00). 3. The PTU Simulator receives PTU_FileReq with file index =3 and request= FR_OPEN_TO_SEND (0) (12 xx xx xx 03 00 00 03 00 00 00 00 00 00 00 00 00 xx xx). 4. The PTU Simulator sends PTU_FileStat with file status FS_XFER_UNDERWAY(value 4) back to the PTU (01 22 xx xx xx 04 04 00 00 00 00 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 xx xx xx xx 00 00 xx xx 00). 5. The PTU Simulator receives NextPackReq from the PTU (06 xx xx xx 20 xx xx). 6. The PTU Simulator sends NextPack to the PTU (01 C6 xx xx xx 21 […]) 7. The PTU Simulator receives NextPackReq from the PTU (06 xx xx xx 20 xx xx). 8. The PTU Simulator sends NextPack to the		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
			PTU (01 C6 xx xx xx 21 […]) 9. The PTU Simulator receives NextPackReq from the PTU (06 xx xx xx 20 xx xx). 10. This process continues until the PTU Simulator sends LastPack to the PTU (01 xx xx xx xx 22 […]). 11. The PTU Simulator receives PTU_FileReq with file index =3 and request= FR_CLOSE_FILE(2) (12 xx xx xx 03 02 00 03 00 00 00 00 00 00 00 00 00 xx xx). 12. The PTU Simulator sends PTU_FileStat with file status FS_XFER_OK(value 4) back to the PTU (01 22 xx xx xx 04 00 00 00 00 00 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 xx xx xx xx 00 00 xx xx 00).		
2	TC3.22.2 Log File Download	See Test Case #1 [Log file download request].	Verify that the PTU Simulator communication logs (step 3 to 12) created in the Test Case #1 [Log file download request] are compliant with Appendix A, Figure A-13.		
3	TC3.22.3 Log File Storing and Display	In the PTU, open the PTU event log created in the Test Case #1 [Log file download request]. In Maintenance menu, select Log View. Verify the DCU logs are similar to Appendix A, Figure A-4, with time stamps, event codes and attributes as shown in the figure.	All event records in the event log simulation file are displayed in the DCU Log tables.		
4	TC3.22.4 Log Database Sorting	1) Click Log table header Event Code column. 2) Click Log table header Event Code column again. 3) Do the same test for all Log table columns.	1) The log records change their display order based on the Event Code ascending number. 2) The log records change their display order based on the Event Code descending number. 3) The log records change their display order from ascending to descending accordingly to the column results.		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
5	TC3.22.5 Log Save	1) Click the Save button on the Log window. On the Save As window, click Save. The Log is saved in the **C:\DenverPTU\DATA\DOOR** folder using the following naming convention: LOG_<Car#>_DOOR_<Location code defined by the supplier>_<Log type defined by the supplier>_<MM-DD-YYYY (date when the log data has been downloaded from the DCU)>_<HH-MM-SS (time when the log data has been downloaded from the DCU)>_<Supplier discretionary data if needed>.TXT 2) Locate and open the saved Log file.	1) The Log is saved. 2) The file contains the following information for each Log event: <Car#>, “DOOR”, <location code defined by the supplier>, <Event sequential number – 4 characters>, <MM-DD-YYYY (date of the event)>, <HH:MM:SS (time of the event in 24-hour format)>, <Event severity code F/W/E/O - character>, <Event unique identification code>, <Event description>, <Supplier discretionary data>.		
6	TC3.22.6 Log Load	1) Continuing from the last test case, close out the PTU Application, and then restart the PTU Application. 2) Click on the LOG button on the PTU Control to launch the DCU Log window, and then click the LOAD button to open up a dialog box for opening up a saved file. 3) Select the file saved in Test Case #5, Log Save, above and open for display in the DCU Log window	1. PTU Application closes out, and then restarts. 2. The DCU Log window opens up and then the dialog box for opening up a saved file opens up. 3. The DCU logs are similar to Appendix A, Figure A-4, with time stamps, event codes and attributes as shown in the figure.		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
7	TC3.22.7 Log File Download Communication Error	1) On the PTU toolbar, click Log Download button. 2) During the download, in the PTU Simulator, click the TX label to disable CRC. 3) Wait until an error message appears. 4) Close PTU Application and do not save the project. In the PTU Simulator, click TX label to enable CRC for a later test.	1) The PTU starts to download the event log file. 2) The PTU Simulator TX text color becomes red. 3) Message bubble appears because the CRC is disabled. The error message disappears. 4) The PTU main window is closed. The PTU Simulator TX text color becomes black.		
8	TC3.22.8 Log File Delete Request	Do Test Case #1 of section 5.3.9 [Versions request from PTU]. In the PTU, click Administration → Log Erasure.	The Delete Event Log File confirmation window pops up.		
9	TC3.22.9 Log File Delete Confirmation Asked	1) Click Cancel button. 2) In the PTU Simulator, compose PTU_FileDeleteStatus in channel C1 with data 03 and enable to send to the PTU. In the PTU, click Maintenance → Log Erasure. 3) Check the **checkbox** then click **Erase Log**. 4) Click Yes. 5) Click “OK” to close the message window. Close PTU Application and do not save the project.	1) The confirmation window is closed and no command is sent out. 2) The Delete Event Log File confirmation window pops up. 3) Another confirmation window appears. 4) The PTU confirms that the event log has been deleted. 5) The delete confirmation window is closed. The PTU main window is closed.		
10	TC3.22.10 Log File Delete Confirmation from DCU	See Test Case #9 [Log file delete confirmation asked].			
11	TC3.22.11 Snapshot File Download Request	In the PTU Simulator, click New button to remove all communication logs. In the PTU Simulator, open the DCU snapshot simulation file	The PTU starts the snapshot files download and a progress bar is displayed. After the Snapshot Download is finished, the 20 snapshots are		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
		snapshotL6.tmp for sending (it is located in the same directory as PTU Simulator software). Do Test Case #1 of section 5.3.9 [Versions request from PTU]. Select Snapshot Download from Maintenance -> Other Functions menu.	displayed in the DCU Snapshot File table. Verify that the following steps appeared in the PTU Simulator communication logs for DCU: 1. The PTU Simulator receives PTU_FileReq command with fileindex =6 and request= FR_DIRECTORY_LOOKUP (4) (12 xx xx xx 03 04 00 06 00 00 00 00 00 00 00 00 00 xx xx). 2. The PTU Simulator sends PTU_FileStat with file size and select_file back to PTU (01 22 xx xx xx 04 01 00 00 00 00 06 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 02 B5 00 00 xx xx 00). 3. The PTU Simulator receives PTU_FileReq command with fileindex =6 and request= FR_OPEN_TO_SEND (0) (12 xx xx xx 03 00 00 06 00 00 00 00 00 00 00 00 00 xx xx). 4. The PTU Simulator sends PTU_FileStat with file status FS_XFER_UNDERWAY(value 4) back to the PTU (01 22 xx xx xx 04 04 00 00 00 00 06 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 xx xx xx xx 00 00 xx xx 00). 5. The PTU Simulator receives NextPackReq from the PTU (06 xx xx xx 20 xx xx). 6. The PTU Simulator sends NextPack to the PTU (01 C6 xx xx xx 21 […]) 7. The cycle continues until the PTU Simulator sends LastPack to the PTU (01 xx xx xx xx 22 […]). 8. The PTU Simulator receives PTU_FileReq command with fileindex =6 and request= FR_CLOSE_FILE (2) (12 xx xx xx 03 02 00 06 00 00 00 00 00 00 00 00 00 xx xx). 9. The PTU Simulator sends PTU_FileStat with FS_XFER_OK command (01 22 xx xx xx 04 00 00 00 00 00 06 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 02 B5 00 00 xx xx 00).		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
			10. After the snapshot #1 file transfer is finished, the PTU starts to download snapshot #2 from PTU Simulator. The download process is repeated with different fileindex (7) like download snapshot #1 (steps 1 to 9). 11. The sequence is repeated up to snapshot #20 (index 25). 12. After the snapshot #20 file transfer is finished, the 20 snapshot file headers are displayed in the snapshot file table in the Snapshot window.		
12	TC3.22.12 Snapshot File Download	See step 3 to 9 in the Test Case #11 [Snapshot file download request]	Verify the PTU Simulator communication logs for the Test Case #11 [Snapshot file download request] compliant with Appendix A, Figure A-13.		
13	TC3.22.13 Snapshot File Display	1) Verify 20 snapshot files in DCU Snapshot File table (See Appendix A, Figure A-5). 2) Double-click the snapshot file item #1. 3) Check the first parameter. 4) Verify that the snapshot graphic complies with the Snapshot Simulation data file described in Appendix A, Figure A-6.	1) All Snapshot records in the Snapshot file are displayed in the Snapshot table. 2) The Snapshot view window pops up. 3) The snapshot data is displayed in graphic on window. 4) Snapshot graphic is displayed.		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
14	TC3.22.14 Snapshot File Download Communication Error	1) Select Snapshot Download from Maintenance -> Other Functions menu. 2) During the file transfer, in the PTU Simulator, click TX label to disable CRC. 3) Wait until the error message bubble appears. 4) Click “OK” to close message window. 5) Close PTU Application and do not save the project. In the PTU Simulator, click TX label to enable CRC for a later test.	1) The PTU Simulator TX text becomes red. 2) The PTU start to download all snapshot files. 3) Message bubble “The file transfer is interrupted due to communication error” pops up and the file transfer is stopped because CRC is disabled. 4) The error message is closed. 5) The PTU main window is closed. The PTU Simulator TX text becomes black.		
15	TC3.22.15 Snapshot File Delete Request	Do Test Case #1 of section 5.3.9 [Versions request from PTU]. Select Delete Snapshot from Maintenance -> Other Functions menu.	The Delete Snapshot window pops up.		
16	TC3.22.16 Snapshot File Delete Confirmation by User	1) Select DCU snapshot file #1 to be deleted. Click “Delete” button. 2) Click “No” button. 3) Select DCU snapshot file #1 to be deleted. Click “Delete” button. 4) In the PTU Simulator, compose PTU_FileDeleteStatus with data 06 (snapshot #1) in the channel C1 and enable to send to the PTU. In PTU, click “Yes” button. 5) Close Delete Snapshot window. Close PTU Application and do not save the project.	1) The Delete Snapshot confirmation window pops up. 2) The confirmation window is closed and no command is sent out. 3) The Delete Snapshot confirmation window pops up. 4) The confirmation window is closed. The PTU Simulator receives PTU_FileDeleteReq (07 xx xx xx 45 06 xx xx) command for selected snapshot file #1. The DCU snapshot #1 status on the Delete Snapshot window is changed to “Deleted”. 5) The Delete Snapshot window is closed. The PTU main window is closed.		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
17	TC3.22.17 Snapshot File Delete Confirmation from DCU	See Test Case #16 [Snapshot file delete confirmation by user].			

5.3.24 TC3.23 Data and File Management

Special Procedural Requirements: None.
Measurements: Self-explanatory.

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
1	TC3.23.1 Save Data File	1) Do Test Case #1 of section 5.3.9 [Versions request from PTU]. In the PTU Simulator, open event log simulation file event.log for sending. In the PTU, select Log Download from Maintenance -> Other Functions menu. 2) In the PTU Simulator, open snapshot simulation file snapshot.lss for sending. In the PTU, select Snapshot Download from Maintenance -> Other Functions menu. 3) In the PTU Simulator, open graph simulation file graph.bin for sending. In the PTU, select Graph Download from Maintenance -> Other Functions menu. 4) From PTU File menu, select Save Project as. 5) Enter a file name for PTU data file. Click “Save” button. 6) Open the PTU working directory (C:\DENVERPTU\DATA\DOOR) and verify that the PTU data file with the user given name is present. 7) Close PTU Application.	1) The event log file is downloaded and displayed. 2) The PTU starts to download snapshot file. After the file transfer is finished, 20 snapshots are displayed on the PTU. 3) The PTU starts to download graph files. After the file transfer is finished, the graph is displayed on the PTU. 4) The Save As window pops up. 5) The PTU data will be saved to a file with the user given name under PTU working directory. 6) The PTU data file is present. 7) The PTU main window is closed.		
2	TC3.23.2 Open Data File	1) Restart PTU Application. From PTU File menu, select Open Project command. 2) Enter the file name that you just saved in the Test Case #1 [Save data file]. Click Open button.	1) The PTU main window pops up. The Open window pops up. 2) The PTU Control Menu is opened.		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
3	TC3.23.3 Display Data File	1) Verify the software versions by clicking Maintenance → Read System Software Version. 2) Click Download Event Log button on the PTU Control Menu. Verify that all event log records are shown. 3) Click SNAPSHOT VIEW button in the PTU Control Menu MAINTENANCE, OTHER FUNCTIONS section. Verify snapshot records in the PTU Snapshot File table. 4) Click snapshot item on Snapshot file table to open snapshot view. Verify snapshot contents. 5) Click GRAPH VIEW button on the PTU Control Menu MAINTENANCE, OTHER FUNCTIONS. Verify all door movement graphs.	1) Compare with results from the Test Case #2 of section 5.3.9 [Versions retrieved by PTU]. 2) The Event Log is displayed in the Log table. The log is similar to Figure A-4. 3) The Snapshot file table is displayed with 20 snapshots for each side. 4) The snapshot view window pops up. The snapshot is similar to Figure A-6. 5) The Door Graph window is displayed. The graph is similar to Figure A-10.		
4	TC3.23.4 Disk Error while Saving Data File	1) From PTU File menu, select Save Project as. 2) Select a write protected folder. Click “Save” button. 3) Click “OK” to close message window.	1) The Save As window pops up. 2) An error message pops up. 3) The error message is closed.		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
5	TC3.23.5 Invalid Data File Name	1) Restart PTU Application. From PTU File menu, select Open Project… 2) Enter a file name that does not exist on the disk. Click “Open” button. 3) Click “OK” button. 4) Enter a non PTU data file name. Click “Open” button. 5) Click “OK” button. 6) Create a temporary text file on the desktop and name it “xxx.dat”. In the PTU, select the temporary file. Click “Open” button. 7) Click “OK” button. 8) Close PTU Application.	1) The Open window pops up. 2) “File <Denver> does not exist. Verify that correct file name was given.” message pops up. 3) The message is closed and the Open window pops up again. 4) An error message pops up. 5) The error message is closed and the Open window pops up again. 6) An error message pops up. 7) The message is closed 8) The PTU main window is closed.		

5.3.25 TC3.24 Upgrade of DCU Application Executable Code

Special Procedural Requirements: On the PTU computer, create the subdirectory “C:\DENVERPTU\SWDownload\DOOR” and copy the DCU Application bin file, from the Intranet.
Make sure the PTU Access is in administrator.

Measurements: Self-explanatory.

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
1	TC3.24.1 Application Upgrade Request	1) Make sure the PTU Access is in administrator. In the PTU File Menu, select Properties. 2) In Login Option, select Boot Loader and click OK button. 3) In the PTU Simulator, compose: 1. PTU_Login (2) 2. PTU_Location_ID 3. PTU_SciVersion 4. PTU_Version (00 10 00 07) and enable to send it back to the PTU. In the PTU Main window status bar (at the bottom), click on the disconnected icon ( ). 4) In PTU Control Menu, click SW DOWNLOAD. 5) Open DCU Application file 2005L0091-00RXX.bin. Do not click the “Download” button yet. 6) In the PTU Simulator, open file **App.bin** for receiving (save it in the same directory as the PTU Simulator Application). In the PTU Simulator, compose the following commands in Channel C1 and enable to send them back to the PTU Application: 1. PTU_Logout (4)	1) The PTU Properties window pops up. 2) The PTU Properties window is closed. 3) In the PTU status bar, the disconnected icon is changed to the connected icon ( ). 4) The Open window pops up. The default directory is C:\DENVERPTU\SWDownload\DOOR. 5) The DCU Application is loaded into memory and the file CRC is verified. 6) The DCU Upgrade confirmation window pops up. And the PTU Simulator receives the PTU_Logout (xx 72 01 xx xx) to logout and reboot to Boot Loader.		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
		2. PTU_Login (2) 3. PTU_SciVersion 4. PTU_Logout (2) Switch back to the PTU Application and **click** the "**download**" button to start downloading DCU Application file 2011L0051-00RXX.bin. 7) Minimize PTU Simulator window to accelerate file transfer by avoiding display refresh. In PTU, click **Yes button**. 8) Click "**OK**" to close message window. 9) Verify that the upgrade time took less than 30 minutes. 10) Copy the original DCU Application file (*.bin) from the PTU computer to the PTU Simulator directory. Open a Command Prompt window. In the PTU Simulator directory, type "FC /b App.bin 2011L0051-00RXX.bin".	7) The confirmation window is closed. The PTU Simulator starts receiving DCU Application binary file. 8) The message window is closed. No error occurred. 9) Application upgrade took less than 30 minutes. 10) Verify that both files are identical. The sentence "No differences encountered" should appear.		
2	TC3.24.2 Application Upgrade File Transfer	Verify if the PTU Simulator communication log for Test Case #1 [Application upload request] compliant with Appendix A, Figure A-12	The following operations are found in the log for each upload to the DCU: 1. The PTU sends PTU_FileReq with file index =0 and request=FR_OPEN_TO_RECEIVE (1) (12 xx xx xx 03 01 00 00 00 00 00 00 00 00 00 00 00 xx xx). 2. The PTU Simulator sends PTU_FileStat with file status FS_XFER_UNDERWAY(value 4) back to the PTU (01 22 xx xx xx 04 04 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 xx xx 00). 3. The PTU sends PTU_NextPack with file data to the PTU Simulator. 4. The PTU Simulator sends PTU_NextPackReq to the PTU (01 06 xx xx xx 20 xx xx 00). 5. The step 3 and 4 are repeated until the file is finished.		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
			6. After the PTU Simulator receives PTU_LastPack, it sends PTU_FileXferOver (01 07 xx xx xx 23 01 xx xx 00). 7. The PTU sends PTU_FileReq with file index =0 and request=FR_CLOSE_FILE(2) (12 xx xx xx 03 02 00 00 00 00 00 00 00 00 00 00 00 xx xx). 8. The PTU Simulator sends PTU_FileStat with FS_XFER_OK (0) to the PTU (01 22 xx xx xx 04 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 xx xx 00)		
3	TC3.24.3 Application Upgrade Process Monitoring	See Test Case #1 [Application upload request]	Verify that the progress of the upload is displayed.		
4	TC3.24.4 Application Upgrade Interrupted	1) In PTU Control Menu, click SW DOWNLOAD button. Enter DCU Application file 2011L0051-00RXX. Click “Download” to close message window. 2) Click Yes button. 3) Click the PTU Stop button on the Toolbar. 4) Click “OK” to close the message window.	1) The DCU Upgrade confirmation window pops up. 2) The confirmation window is closed. The PTU Simulator starts to receive DCU App. binary file 3) Message bubble “The file transfer is interrupted due to communication error” appears and the file transfer is stopped. 4) The message is closed.		
5	TC3.24.5 Application Upgrade Communication Error	1) In PTU Control Menu, click SW DOWNLOAD button. Enter DCU Application file 2011L0051-00RXX.bin. Click “Download” to close message window. 2) Click Yes button. 3) Click the PTU Simulator TX label to disable CRC.	1) The DCU Upgrade confirmation window pops up. 2) The confirmation window is closed. The PTU Simulator starts to receive DCU App. binary file 3) The PTU Simulator TX text color becomes red.		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
		4) Wait until the error message appears. 5) Close PTU Application. In the PTU Simulator, click TX label to enable CRC for a later test.	4) Message bubble “The file transfer is interrupted due to communication error” appears and the file transfer is stopped. The error message disappears. 5) The PTU main window is closed. The PTU Simulator TX text color becomes black.		
6	TC3.24.6 Physical Disconnection from DCU During Application Upgrading	1) Do steps 1 to 6 of test case #1 Section 5.3.25 2) Click Yes button. 3) Disconnect the PTU from the RS-232 port of the PTU computer. 4) Click OK to close the message window. 5) Close PTU Application. Reconnect the PTU cable to the PTU computer.	1) The Open window pops up. The DCU App. is loaded into memory and the file CRC is verified. The DCU Upgrade confirmation window pops up. 2) The confirmation window is closed. The PTU Simulator starts to receive DCU App. binary file. 3) “The file transfer is interrupted due to communication error” message pops up and the file transfer is stopped. 4) The error message is closed. 5) The PTU main window is closed.		

5.3.26 TC3.25 Upgrade of DCU Boot Loader

Special Procedural Requirements: On the PTU computer, copy the DCU Boot Loader H86 file (2011L0053-00RXX.h86) from the Intranet in the "C:\DENVERPTU\SWDownload\DOOR" subdirectory.
Make sure the PTU Access is in administrator.
Make sure the PTU computer is connected to the DCU and the power is ON.
Make sure the DCU Controller that will be programmed is in Bootstrap mode.

Measurements: Self-explanatory.

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
1	TC3.25.1 DCU Boot Loader upgrade Request	1) In PTU Control Menu, click OTHER FUNCTIONS button under ADMINISTRATION and then LOAD FLASH->OPEN FLASH FILE button. Open Boot Loader file 2011L0053-00RXX.h86. 2) In PTU Control Menu, click OTHER FUNCTIONS button under ADMINISTRATION and then LOAD FLASH-> SEND FLASH FILE button. 3) Follow the prompt of the PTU Application. 4) Follow the prompt of the PTU Application, and verify that the uprade took less than 30 minutes. 5) In the PTU Control Menu, click on SYSTEM SW VERSION button under MAINTENANCE and verify that the Boot Loader Version and Date are the same as the file downloaded at step 1. 6) In the PTU menu, select Tools -> DCU Code Status and verify that the DCU Boot Loader Size and CRC are the same as the file downloaded at step 1.	1) The Open window pops up. The default directory is C:\DENVERPTU\SwDownload\DOOR. 2) A message window pops up requesting to set the controller to program in Bootstrap mode and then to reboot the DCU. 3) The download process is started and status is displayed. A message window pops up requesting to remove the Bootstrap mode and reset the DCU. 4) The DCU is reset and the PTU automatically logs into the DCU. And upgrade time is less than 30mins. 5) The System SW Version window pops up. The DCU Boot Loader Version and Date are the same as the file downloaded at step 1. 6) The DCU Code Status window pops up. The DCU Boot Loader Size and CRC are the same as the file downloaded at step 1.		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
		7) Close the PTU application and do not save the project.	7) The PTU main window is closed.		
2	TC3.25.2 RTP Tool Upgrade request	1) Repeat the previous test case with a RTP tool file. 2) Redo the Boot Loader Upgrade to replace the RTP Tool	1) The RTP Tool is uploaded correctly 2) The Boot Loader is uploaded correctly.		

5.3.27 TC3.26 Upgrade of DCU Neuron Application

Special Procedural Requirements: On the PTU computer, copy the DCU Neuron Gateway Application nxe file (2011L0052-00RXX.nxe) from the Intranet in the "C:\DENVERPTU\SWDownload\DOOR" subdirectory.
Make sure the PTU Access is in administrator.
Make sure the PTU computer is connected to the DCU and the power is ON.

Measurements: Self-explanatory.

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
1	TC3.26.1 DCU Neuron Application upgrade Request	1) Start PTU Application. In PTU Control Menu, click OTHER FUNCTIONS button under ADMINISTRATION and then NODEUTIL button. 2) Send the Service Pin. 3) On the NodeUtil window, type G to go to device menu. Type 1 and press [Enter] to choose node 1. 4) Type Y to download Neuron executable. Type the full path and the name of the file to program and press [Enter]. 5) Type Y to continue, then type E twice to exit NodeUtil. 6) Close the PTU application and do not save the project.	1) The NodeUtil window is open. 2) The message "Received an ID message from device 1. Program ID is DCU, Vapor" is displayed. 3) The device menu is displayed. 4) The download process is started. When the file is successfully downloaded, the message "Do you want to set the node configured, on-line?" is displayed. 5) NodeUtil window is closed. 6) The PTU main window is closed.		

5.3.28 TC3.27 PTU Security

Special Procedural Requirements: Make sure the PTU Access is in administrator.
Measurements: Self-explanatory.

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
1	TC3.27.1 PTU Password	1) Start PTU. 2) On the PTU Application, unlock administration functions: go to menu Administration → Other Functions → Password, write **vsrsdenver** and press OK.	1) The “Access: User” message is displayed in the PTU Control Menu. 2) The “Access: Admin” message is displayed in the PTU Control Menu.		
2	TC3.27.2 DCU Application Upgrade, Flash Programming, Neuron Application Upgrade, DCU Configuration, Snapshot Erasure, DCU Log and System Counters resetting are available when PTU Access is administrator	1) Do steps 1 to 5 of test case #8 of section 5.3.4 [DCU Application sends a valid Login]. 2) In PTU Administration menu, verify that the SW Download is activated. 3) In PTU Administration -> Other Functions menu, verify that the DCU Configure, LoadFlash -> Open Flash File and NodeUtil items are activated. 4) In PTU, select DCU Configure from the Administration -> Other Functions menu. Verify that the Apply button is activated. 5) In PTU Control menu, verify that the Log Erasure button under Administration->Other Functions group is activated, as are the System Counters resetting functions, and Snapshot Erasure buttons under Maintenance-> Other Functions->System Counter group. 6) Restart PTU. Do steps 1 to 5 of test case #8 of section 5.3.4 [DCU Application sends a valid Login].	1) The DCU is logged in. 2) The SW Download File menu item is activated (not grayed out). 3) The Open Flash File and NodeUtil items are activated (not grayed out). 4) The Apply button is activated (not grayed out). 5) The “Log Erasure” button is activated (not grayed out), as are the System Counters resetting functions, and Snapshot Erasure buttons within the Other Functions group. 6) The DCU is logged in.		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
		7) In PTU Administration menu, verify that the SW Download item is deactivated. 8) In PTU Administration -> Other Functions menu, verify that the LoadFlash -> Open Flash File item is deactivated. 9) In PTU Administration -> Other Functions file menu, verify that the NodeUtil (for Neuron Upgrade) item is deactivated. 10) In PTU, select DCU Configure from the Administration -> Other Functions menu. Verify that the Apply button is deactivated. 11) In PTU Control menu, verify that the Log Erasure button under Administration group is deactivated, as are the System Counters resetting functions, and Snapshot Erasure buttons within the Other Functions group.	7) SW Download menu item is deactivated (grayed out). 8) The Open Flash File menu item is deactivated (grayed out). 9) The NodeUtil (for Neuron Upgrade) menu item is deactivated (grayed out). 10) The Apply button is deactivated (grayed out). 11) The “Log Erasure” button is deactivated (grayed out), as are the System Counters resetting functions, and Snapshot Erasure buttons within the Other Functions group.		

5.3.29 TC3.28 Help File

Special Procedural Requirements: None.
Measurements: Self-explanatory.

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
1	TC3.28.1 Display Help Information	1) Start PTU Application. 2) In the PTU Help menu, select Help Contents. 3) Close the PTU Software User Manual window. Close PTU Application.	1) The PTU main window pops up. 2) The PTU Application Software User Manual is opened. 3) The PTU Software User Manual window is closed. The PTU main window is closed.		

5.3.30 TC3.29 Printing

Special Procedural Requirements: None.
Measurements: Self-explanatory.

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
1	TC3.29.1 Print Selected Log and Snapshot Files	1) Start PTU Application. 2) From the PTU File menu, Open the PTU data file saved in Test Case #1 of section 5.3.24 [Save data file]. In the PTU File menu, select Print 3) Select the printer. Click Print button on the Print window. 4) Close PTU Application and do not save the project.	1) The PTU Control Menu pops up. 2) The Print window pops up. 3) The selected PTU data are sent to printer to print out. 4) The PTU main window is closed.		

5.3.31 TC3.30 System Counters

Special Procedural Requirements: Make sure the PTU Access is in administrator.
Measurements: Self-explanatory.

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
1	TC3.30.1 Request DCU System Counters and Display.	1) Do Test Case #1 of section 5.3.9 [Versions request from PTU]. In the PTU Simulator, open the simulation file SystemCounters.tmp for sending (in the directory of the PTU Simulator). Click MAINTENANCE -> OTHER FUNCTIONS -> SYSTEM COUNTERS VIEW button in the PTU Control Menu. 2) Click Download button in the window. Compare the figure with Figure A-11.	1) The System Counters window pops up. 2) The PTU Simulator starts file transferring, with a progress bar displaying the status of the transfer. After the file transfer is completed, the System Counters are displayed in the System Counters window.		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
2	TC3.30.2 System Counters Download	Verify if the PTU Simulator communication log for Test Case #1 [Request DCU System Counters and Display] is compliant with Appendix A, Figure A-13.	The following operations are found in the log: 1. The PTU sends PTU_FileReq with file index =28 and request=FR_OPEN_TO_SEND (0) (12 xx xx xx 03 00 00 1C 00 00 00 00 00 00 00 00 00 xx xx). 2. The PTU Simulator sends PTU_FileStat with file status FS_XFER_UNDERWAY(value 4) back to the PTU (01 22 xx xx xx 04 04 00 00 00 00 1C 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 xx xx xx xx 00 00 xx xx 00). 3. The PTU sends PTU_NextPackReq to the PTU Simulator (06 xx xx xx 20 xx xx). 4. The PTU Simulator sends PTU_NextPack with file data to the PTU. 5. The step 3 and 4 are repeated until the file is finished. 6. After the PTU receives PTU_LastPack, it sends PTU_FileReq with file index =28 and request=FR_CLOSE_FILE(2) (12 xx xx xx 03 02 00 1C 00 00 00 00 00 00 00 00 00 xx xx). 7. The PTU Simulator sends PTU_FileStat with FS_XFER_OK (0) to the PTU (01 22 xx xx xx 04 00 00 00 00 00 1C 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 xx xx xx xx 00 00 xx xx 00).		
3	TC3.30.3 Reset System Counters	1) Connect the PTU Application to the DCU. 2) In the PTU Application, click on the Maintenance→Other Functions→System Counters Download on the PTU Control Menu.	1) The PTU Application logins to the DCU. 2) The PTU Application downloads the counter from the DCU and the System Counters window is displayed. Door Cycles: _____ (A) Last Door Cycles Reset Date: _____ (B) Last Door Cycles: _____ Previous Door Cycles Reset Date: _____		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
			Operating Hours: _____ (C) Last Operating Hours Reset Date: _____ (D) Last Operating Hours: _____ Previous Operating Hours Reset Date: _____ Last Event Occurences Reset Date: _____ (E) Previous Event Occurences Reset Date: _____		
		3) In the System Counters window, click on the Reset Cycle Counter button. Click on Yes button to confirm the reset.	3) Followings are displayed: Door Cycles: 0 Last Door Cyles Reset Date: [NOW] Last Door cycles: Same as (A) Previous Door Cycles Reset Date: Same as (B)		
		4) In the System Counters window, click on the Reset Operating Hours button. Click on Yes button to confirm the reset.	4) Followings are displayed: Operating Hours: 0 Last Operating Hours Reset Date: [NOW] Last Operating Hours: Same as (C) Previous Operating Hours Reset Date: Same as (D)		
		5) In the System Counters window, click on the Reset Event Occurrences button. Click on Yes button to confirm the reset.	5) Followings are displayed: Last Event Occurences Reset Date: [NOW] Previous Event Occurences Reset Date: Same as (E)		

5.3.32 TC3.31 Program Removal

Special Procedural Requirements: None.
Measurements: Self-explanatory.

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
1	TC3.31.1 Uninstall Program Is Registered So That It Is Listed on the Uninstall Page of the Add/Remove Program Windows Utility.	Open Windows Control Panel, select “Add or Remove Programs”. Verify if there is an entry for “Denver PTU: VAPOR Door Version x.y.z”, where x.y.z is the actual version.	The entry is present.		
2	TC3.31.2 Remove PTU Software	1) Select “Denver PTU: VAPOR Door Version x.y.z” and click the “Change/Remove” button. Answer “Yes” when asked for confirmation (NOTE: The PTU Application Removal will automatically invite the user to uninstall the OpenLDV Echelon’s driver). After the removing operation is completed, click on the PTEProgram shortcut on the desktop. 2) Delete the shortcut.	1) The “Application Not Found” error message is displayed. 2) The shortcut is deleted.		
3	TC3.31.3 Uninstall program removes all files in Program Files folder that it uniquely owns.	After the removing operation is completed, verify if the PTU software is removed from the computer by verifying that the PTU installation directory (C:\DENVERPTU\Program\DOOR-Door Subsystem\DOOR-*x.y.z*, where *x.y.z* represents the version of the PTU under test) is automatically removed.	The folder is removed.		
4	TC3.31.4 Uninstall program leaves no remaining file or registry entry other than the files created by the user.	1) Verify that the PTU user data is still present by verifying the PTU data directory (C:\ DENVERPTU \Data\DOOR). 2) Verify if Denver PTU Registry keys are removed. 3) “Non-user” files’ deletion is verified in previous test case.	1) The folder (C:\ DENVERPTU \Data\DOOR) still contains the files created during PTU use. 2) Registry keys created during setup are automatically removed during program removal operation.		

5.3.33 TC3.32 File Structure and Security Restrictions

Special Procedural Requirements: None.
Measurements: Self-explanatory.

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
1	TC3.32.1 PTU Application works for any drive letter (C:, D:, …)	1) Open a command prompt window by clicking on the start menu, selecting “Run…”, typing “cmd”, and pressing [Enter]. On the command prompt window type: “md c:\ptutest”. Type: “subst x: c:\ptutest”. Close the command prompt window by typing “exit”. 2) Start the installation program setup.exe. When choosing the installation directory, replace “C:” by “X:”. Continue the installation, until the end. The PTU is installed on drive X: under \DENVERPTU folder. When the installation is completed, start windows explorer go to X:\DENVERPTU\Program\DOOR-Door Subsystem\DOOR-*x.y.z* (x.y.z is the PTU version). Make sure the PTU works properly by executing step 1 of Test Case #1 from section 5.3.4. 3) The PTU is already tested for drive C: and drive X:, we can assume that the PTU works on any drive letter.	1) Virtual drive X: is created in c:\ptutest folder. 2) The PTU works as expected.		
2	TC3.32.2 All PTU Application generated data are stored in X:\DENVERPTU\DATA\DOOR	Open folder in Windows explorer. Run the PTEProgram located in X:\DENVERPTU\Program\DOOR-Door Subsystem\DOOR-*x.y.z.* Redo Test Case #1 of section 5.3.24.	All files transferred to PTU are stored in X:\DENVERPTU\DATA\DOOR folder.		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
3	TC3.32.3 “Everyone” has read / write access to X:\DENVERPTU\DATA\DOOR	1) Open Windows explorer. Go to X:\DENVERPTU. Right-click on the folder and select Properties. In Security tab, select group “Users” and make sure Modify and Write box are **not** checked. Click OK. Create and right-click on X:\DENVERPTU\Data\DOOR folder and select Properties. In Security tab, select group “Users” and make sure Modify and Write box are not checked. Click OK. 2) Make sure there is at least one restricted user account on the PTU computer. (You can create a user account by going to the Windows Control Panel and selecting User Accounts.) Logout from Windows and login using the restricted user account. 3) Go to X:\DENVERPTU\Programs\ DOOR Door Subsystem\DOOR-*x.y.z* and run PTEProgram.exe. In File menu, select Save Project As. Save the file in X:\DENVERPTU\DATA\DOOR. Close PTU Application. 4) Logout from Windows and login with the administrator account.	1) The rights are set for the PTU Application folders. 2) You are logged into PTU computer with restricted rights. 3) The PTU user is able to save a project in the data folder. 4) You are logged into PTU computer with administrator rights.		
4	TC3.32.4 “Everyone” has read access to X:\DENVERPTU\Program\DOOR-Door Subsystem\DOOR-*x.y.z*	Tested in previous Test Case.			
5	TC3.32.5 Executable program, help file, DLL and any other data that allow PTU Application to work are stored in X:\DENVERPTU\Program\DOOR-Door Subsystem\DOOR-version-note	Go to X:\DENVERPTU\Program\ DOOR-Door Subsystem\DOOR-*x.y.z*. Verify that the following mandatory files are present: PTUPROGRAM.exe, PTU_Application.DLL and NodeUtil.exe.	Files are present		

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
6	TC3.32.6 Software download packages are located in X:\DENVERPTU\SWDownload\DOOR	Tested in section 5.3.25. The default folder when opening a DCU application or a boot loader is X:\DENVERPTU\SWDownload\DOOR.	See section 5.3.25 output.		
7	TC3.32.7 Temporary files are stored in X:\DENVERPTU\Temp and are deleted when PTU application exits	The temp folder is not used by the PTU Application.			

5.3.34 TC3.33 Date and Time Presentation

Special Procedural Requirements: None.
Measurements: Self-explanatory.

#	Test Cases	Input Specifications	Output Specifications	Result	
				P	F
1	TC3.33.1 All dates displayed by the PTU use the form MM-DD-YYYY. Month, day and year use Integer values. Year uses four digits	1) In PTU, open the data file saved in section 5.3.24. 2) Click on the System SW Version button. Close the System SW Version window. 3) Click on the Log button. Verify the date and time format. Close the Log window. 4) Click on Other Functions button, and then on Snapshot button. Click the first snapshot. Close the snapshot window and the snapshot list window. 5) Click on Help menu, and then on About.	1) The data file is open. 2) The System SW Version date format is as expected (MM-DD-YYYY). 3) The Log date and time format is as expected (MM-DD-YYYY HH:MM:SS). 4) The Snapshot date and time format is as expected. 5) The About box date format is as expected.		
2	TC3.33.2 All times displayed by the PTU use the form HH:MM:SS. “HH” is hours from 00 to 23, “MM” is minutes from 00 to 59, and “SS” is seconds from 00 to 59	Tested in previous test case.	See previous test case output.		
3	TC3.33.3 The PTU Application must be able to process all dates in the range of 2000 to 2050	Tested in Test Case #1 of Section 5.3.20.			

Appendix A: Figures and Tables

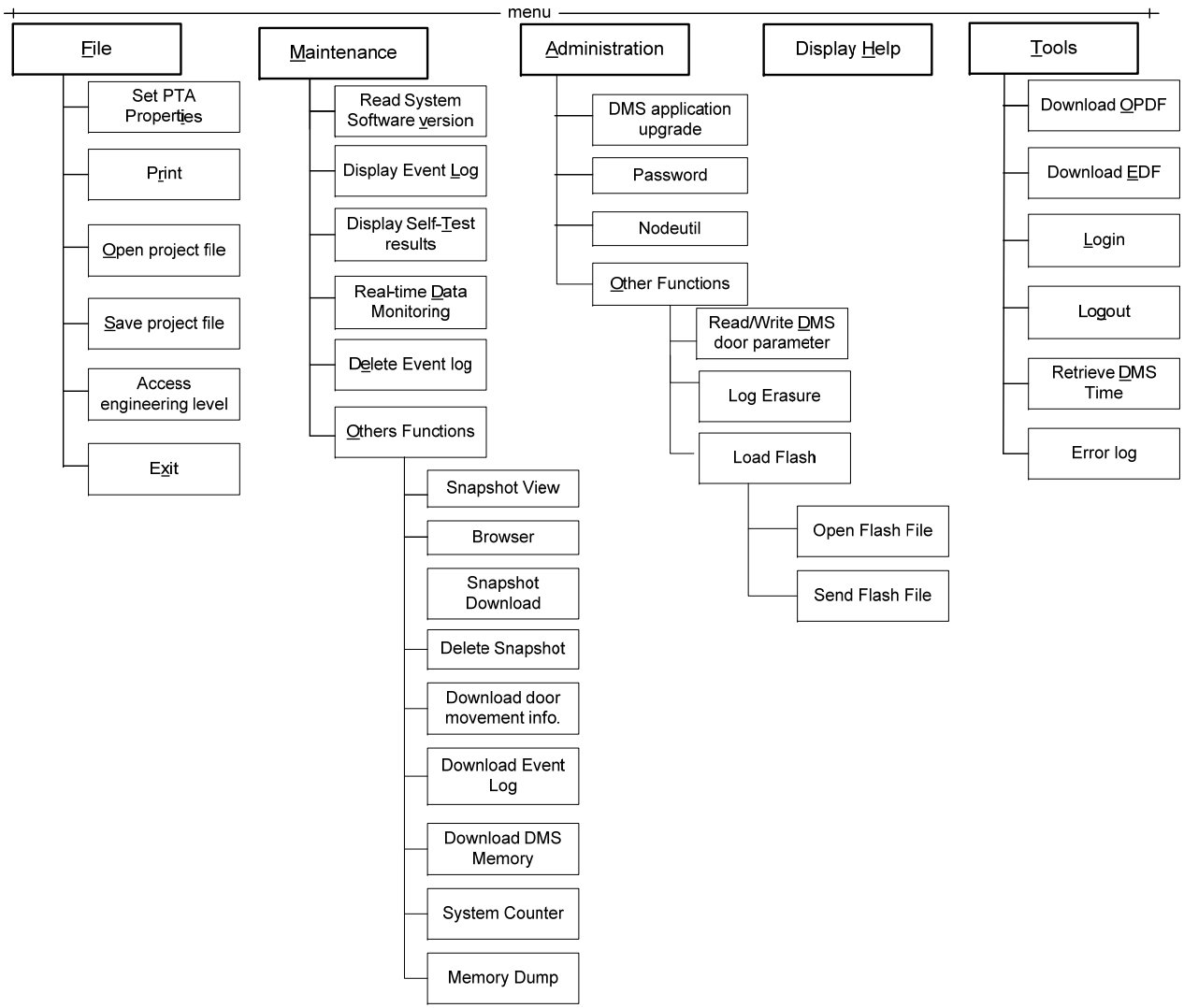

Figure A-3: PTU Application Menu

DCU LOG

* = Time innacurate

Load | Refresh | Save

Index	Date (mm-dd-yyyy)	Time (hh:m:s)	Type	State	Snapshot	Event Code	Description
1	06-09-2008	09:14:47	Pulse			54	Event Log Erased by PTE
2	06-09-2008	09:14:47	Fault/Reset	Fault		7	Motor Power not Available
3	06-09-2008	09:14:47	Fault/Reset	Fault		9	Mechanical Lock Off/Cut-Out
4	06-09-2008	09:14:47	Fault/Reset	Fault		57	Network Communication Lost
5	06-09-2008	09:14:47	Fault/Reset	Fault		60	Invalid Location ID Detected by Neur
6	06-09-2008	09:14:52	Pulse			55	Event Log Retrieved by PTE
7	06-09-2008	*09:15:54	Pulse			58	System Boot-Up
8	06-09-2008	*09:17:16	Pulse			58	System Boot-Up
9	06-09-2008	*09:17:22	Pulse			58	System Boot-Up
10	06-09-2008	*09:17:22	Fault/Reset	Reset		7	Motor Power not Available
11	06-09-2008	*09:17:22	Fault/Reset	Fault		28	Motor Encoder Cable/Circuit Error

Figure A-4: Event Log

Snapshot Table

No.	Time (m-d-y h:m:s)	Event Code	Rate	# of Parameter	Parameters
1	06:09-2008 09:17:22	38 Motor OC Shutd…	5	57	1,2,3,4,5,6,7,8,9
2	06:09-2008 09:17:22	38 Motor OC Shutd…	5	57	1,2,3,4,5,6,7,8,9
3	06:09-2008 09:17:22	38 Motor OC Shutd…	5	57	1,2,3,4,5,6,7,8,9
4	06:09-2008 09:17:22	38 Motor OC Shutd…	5	57	1,2,3,4,5,6,7,8,9
5	06:09-2008 09:17:22	38 Motor OC Shutd…	5	57	1,2,3,4,5,6,7,8,9
6	06:09-2008 09:17:22	38 Motor OC Shutd…	5	57	1,2,3,4,5,6,7,8,9
7	06:09-2008 09:17:22	38 Motor OC Shutd…	5	57	1,2,3,4,5,6,7,8,9
8	06:09-2008 09:17:22	38 Motor OC Shutd…	5	57	1,2,3,4,5,6,7,8,9
9	06:09-2008 09:17:22	38 Motor OC Shutd…	5	57	1,2,3,4,5,6,7,8,9
10	06:09-2008 09:17:22	38 Motor OC Shutd…	5	57	1,2,3,4,5,6,7,8,9
11	06:09-2008 09:17:22	38 Motor OC Shutd…	5	57	1,2,3,4,5,6,7,8,9
12	06:09-2008 09:17:22	38 Motor OC Shutd…	5	57	1,2,3,4,5,6,7,8,9
13	06:09-2008 09:17:22	38 Motor OC Shutd…	5	57	1,2,3,4,5,6,7,8,9
14	06:09-2008 09:17:22	38 Motor OC Shutd…	5	57	1,2,3,4,5,6,7,8,9
15	06:09-2008 09:17:22	38 Motor OC Shutd…	5	57	1,2,3,4,5,6,7,8,9

Figure A-5: Snapshot

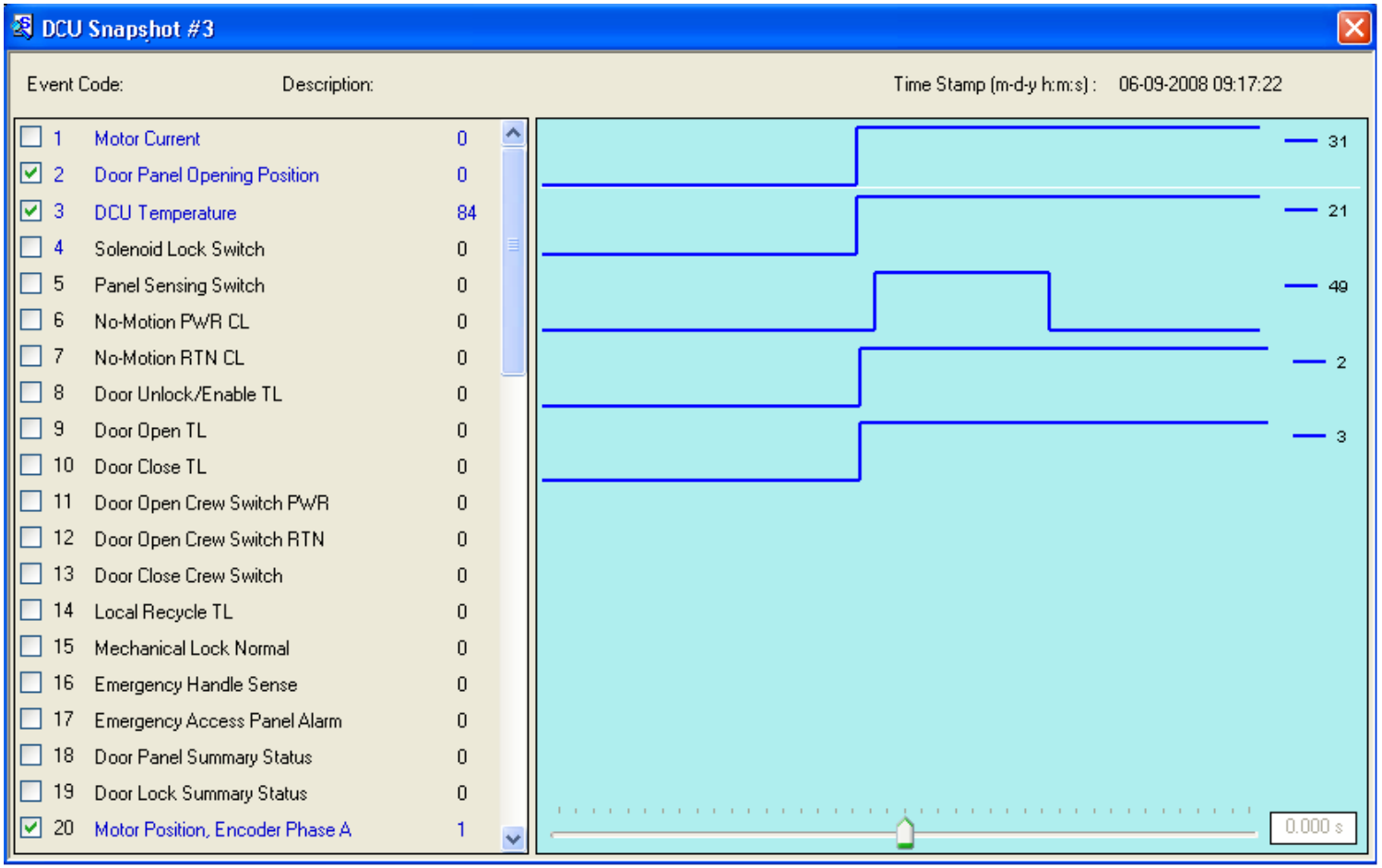

Figure A-6: Snapshot Data In Graphic

Operating Parameter Description File (OPDF)

OPDF Version: 0.7

Parameter_Code	Description	Unit	Sampling_Period
0	Delta MS		10ms
1	Motor Current	A	50ms
2	Door Panel Opening Position	%	50ms
3	DMCU Temperature	°F	50ms
4	Solenoid Lock Switch		50ms
5	Panel Sensing Switch		50ms
6	No-Motion PWR CL		50ms
7	No-Motion RTN CL		50ms
8	Door Unlock/Enable TL		50ms
9	Door Open TL		50ms
10	Door Close TL		50ms
11	Door Open Crew Switch PWR		50ms
12	Door Open Crew Switch RTN		50ms
13	Door Close Crew Switch		50ms

Figure A-7: Operating Parameters List

DATA MONITORING: BROWSER

Para.ID	Description	Unit	Value
1	Motor Current	A	0.0
2	Door Panel Opening Position	%	0
3	DCU Temperature	oF	71
4	Solenoid Lock Switch 1		1
5	Solenoid Lock Switch 2		1
6	Panel Sensing Switch Left		1
7	Panel Sensing Switch Right		1
8	Zero Speed PWR CL		1
9	Zero Speed RTN CL		1
10	Open PWR TL		1
11	Open RTN TL		1
12	Close TL		1
13	Crew Switch Open		1
14	Crew Switch Enable		1
15	Crew Switch Close		1

Figure A-8: Parameters Value

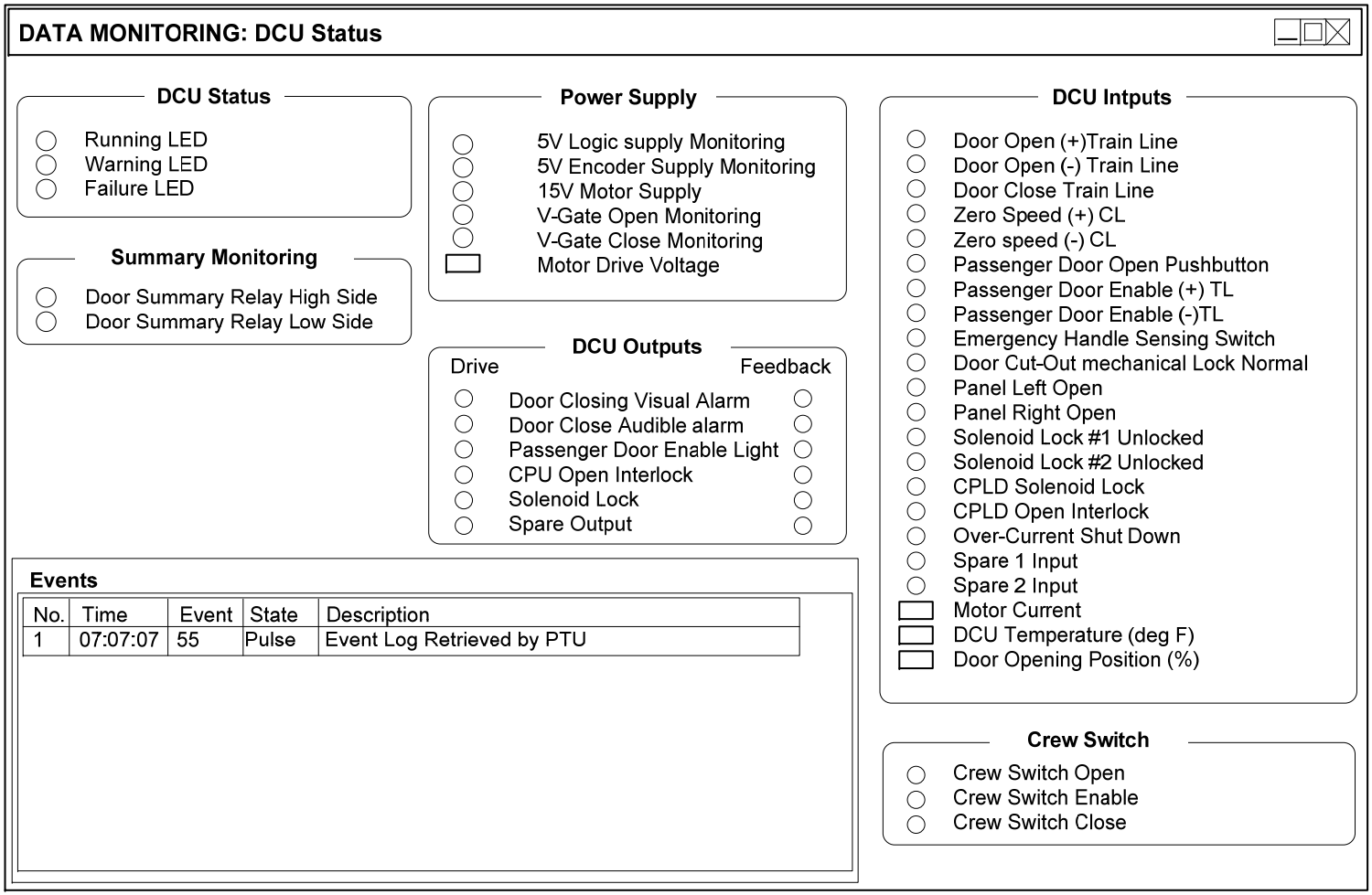

Figure A-9: DCU Status Information Displayed

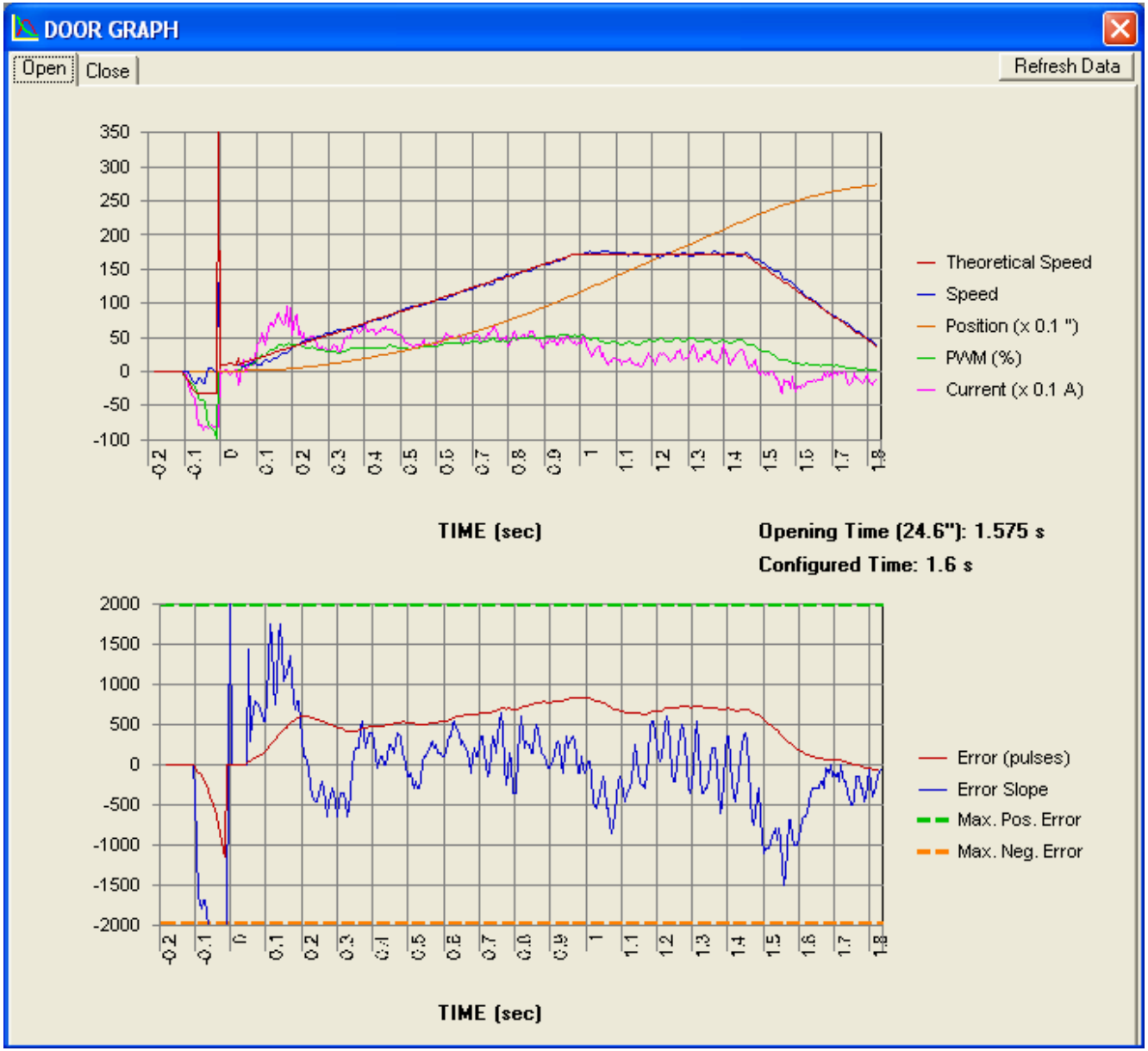

Figure A-10: DCU Door Graph displayed

SYSTEM COUNTERS

Download | Reset Door Cycles | Reset Operating Hours | Reset Event Occurences

#	Value	Description
	0	Door Cycles
	12/8/2007/5:34:12PM	Last Door Cycles Reset Date
	0	Last Door Cycles
	1/1/2007	Previous Door Cycles Reset Date
	0.2	Operating Hours
	12/8/2007/5:34:12PM	Last Operating Hours Reset Date
	0.0	Last Operating Hours
	1/1/2007	Previous Operating Hours Reset Date
	12/8/2007/5:34:12PM	Last Event Occurrences Reset Date
	1/1/2007	Previous Event Occurrences Reset Date
1	0	Watchdog Reset Detected
2	0	DCU App. CRC32 Discrepancy
3	0	Boot Loader CRC32 Discrepancy
4	0	SRAM Failure
5	0	NVRAM Failure
6	1	SPI Error
7	0	Motor Power not Available
8	0	Motor Power Supply Below 30 V

Figure A-11: DCU System Counters window

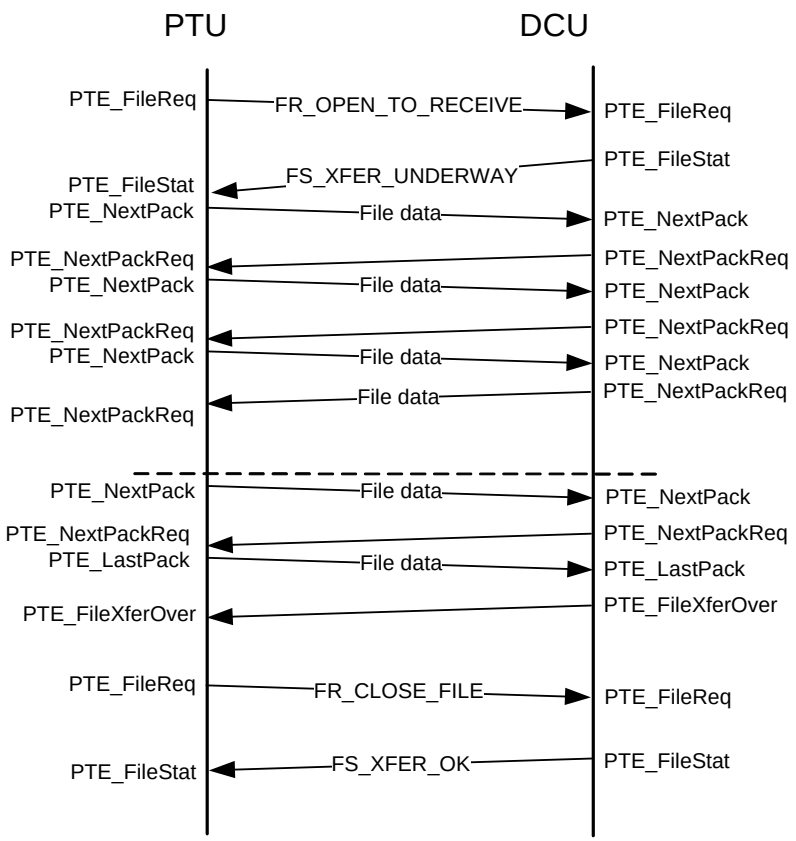

Figure A-12: File Transfer Protocol Implementation – File Upload and No Error

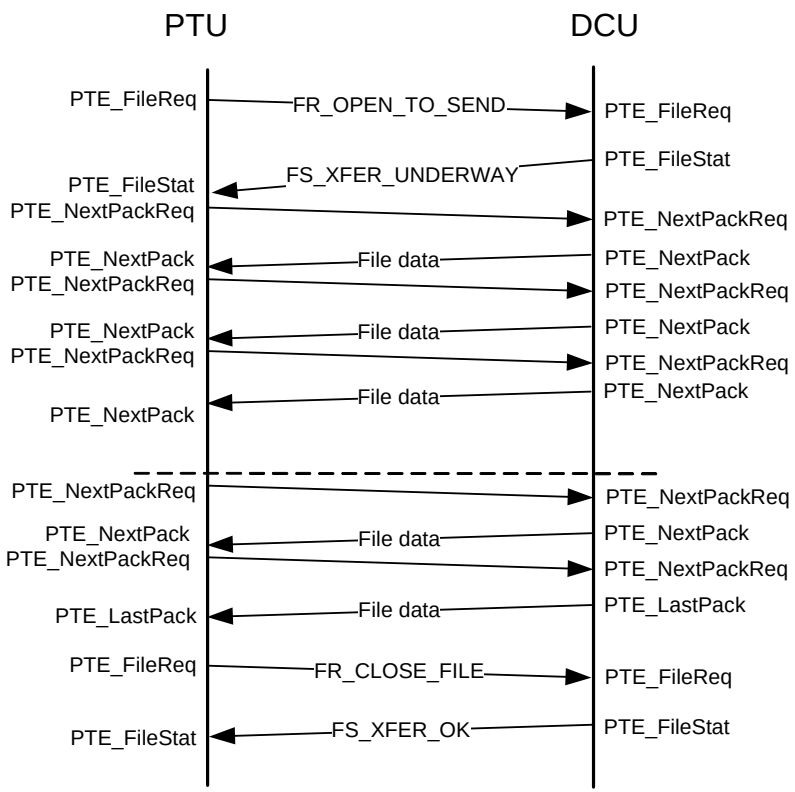

Figure A-13: File Transfer Protocol Implementation – File Download and No Error

Table A-3: Event Description File (Version 0.0)

Event #	Event Name	Type	Event Description
1	Watchdog Reset Detected	P	The DCU logs this event when the processor has been reset by the watchdog timer circuit.
2	DCU App. CRC32 Discrepancy	S/R	The DCU sets this event when the CRC32 of the application does not match with CRC32 value stored in the Application table. The DCU resets this event when the CRC32 of the application matches with CRC32 value stored in the Application table.
3	Boot Loader CRC32 Discrepancy	S/R	The DCU sets this event when the CRC32 of the Boot Loader application does not match with CRC32 value stored in the Boot Loader table. The DCU resets this event when the CRC32 of the Boot Loader application matches with CRC32 value stored in the Boot Loader table.
4	SRAM Failure	S/R	The DCU sets this event when at least one SRAM cell or SRAM addressing does not work properly. The DCU resets this event when the SRAM is fully functional.
5	NVRAM Failure	S/R	The DCU sets this event when at least one NVRAM cell or NVRAM addressing does not work properly. The DCU resets this event when the NVRAM is fully functional.
6	SPI Error	S/R	The DCU sets this event when communication errors repeated on the Neuron SPI link. The DCU resets this event when the communication is working without error.
7	Motor Power not Available	S/R	The DCU sets this event when the Motor Power input voltage is below functional range (<22 V). The DCU resets this event when the Motor Power input voltage is within functional range (22-44 V).
8	Motor Power Supply Below 30 V	S/R	The DCU sets this event when the Motor Power input voltage is between 22 and 30 Volts. The DCU resets this event when the Motor Power returns over 30 Volts.
9	Door Cut-Out Mechanical Lock in Not Normal	S/R	The DCU sets this event when the Door Cut-Out Mechanical Lock is not in normal. The DCU resets this event when the Door Cut-Out Mechanical Lock signal is in normal.
10	-	-	Unused.
11	Emergency Handle Activation	S/R	The DCU sets this event when the Emergency Handle signal is activated. The DCU resets this event when the Emergency Handle signal is deactivated.
12	-	-	Unused.
13	Door Open TLs Mismatch	S/R	The DCU sets this event when there is a mismatch between the Door Open (+) TL and the Door Open (-) TL signals. The DCU resets this event when both Door Open (+) and (-) TLs signals are active. ***Note: The DCU does not set this event if the Door Cut-Out Mechanical Lock is not in Normal.***
14	Door Unlocking Timeout	S/R	The DCU sets this event when the Lock Sensing Switches indicate locked for more than 1.4s after an unlock command. The DCU resets this event when the next time the Lock Sensing Switches indicate unlocked within 1.4s from the unlock command.
15	Door Opening Time Longer than Expected	S/R	The DCU sets this event when the door opening exceeds 1s more than the configured opening time. The DCU resets this event when the door is fully open.
16	Door Opening Timeout	S/R	The DCU sets this event when the door opening exceeds 10s more than the configured opening time. The DCU resets this event when the door is fully open.
17	Loss of “Closed&Locked” Status w/o Door Command	S/R	The DCU sets this event when doors are unlocked or open without an opening command or an emergency activation. The DCU resets this event when doors are closed and locked.
18	Door Closing Time Longer than Expected	S/R	The DCU sets this event when the door closing exceeds 1s more than the configured closing time. The DCU resets this event when doors are closed and locked.
19	Door Locking Time Longer than Expected	S/R	The DCU sets this event when it takes more than 1s from the configured closing time to lock doors. The DCU resets this event when doors are closed and locked.
20	Loss of Zero Speed with Door Open	S/R	The DCU sets this event when any Zero Speed signal is lost with open or unlocked doors. The DCU resets this event when both Zero Speed signals are present or doors are closed and locked.
21	PSS Mismatch	S/R	The DCU sets this event when there is a mismatch between left and right Panel Sensing Switch signals. The DCU resets this event when both Panel Sensing Switch signals are consistent.
22	Door Closing Obstruction	S/R	The DCU logs this event when the DCU is unable to close doors after exceeding the number of configured obstruction retry.
23	Motor Open Supply Error	S/R	The DCU sets this event when the motor open supply signal is active without valid logic for more than 2s. The DCU resets this event when the motor open supply signal is inactive.

Event #	Event Name	Type	Event Description
24	Solenoid Lock Supply Error	S/R	The DCU sets this event when the solenoid lock supply signal is active without a valid logic for more than 0.2s. The DCU resets this event when the solenoid lock supply signal is inactive.
25	-	-	Unused.
26	-	-	Unused.
27	Motor Encoder Phase Lost Error	S/R	The DCU sets this event when there are more than 20 transitions on one encoder phase and no transition on the other phase within 5ms. The DCU resets this event whenever there are transitions on both phases.
28	Motor Encoder Cable/Circuit Error	S/R	The DCU sets this event when encoder phase A, phase B, and Index inputs are all on. The DCU resets this event when any one of the encoder signal is off.
29	Motor Control Module Communication Error	S/R	The DCU sets this event when the motion control module is unable to handle requests from the door operator module. The DCU resets this event when the next time a request is accepted.
30	Motor Drive Error or Wire	S/R	The DCU sets this event when the motor wire is disconnected or the motor does not respond for 0.2s. The DCU resets this event when the motor responds to the DCU command within 0.2s.
31	Motor Drive Circuit (H-Bridge) Error	S/R	The DCU sets this event when the motor current flows with a single active transistor on the motor drive during the test. The DCU resets this event when each transistor passes the test.
32	Motor Current Reading Error	S/R	The DCU sets this event when the motor current reading is out of the expected range. The DCU resets this event when motor current reading is within range.
33	Motor Voltage Reading Error	S/R	The DCU sets this event when the motor voltage reading is out of expected range. The DCU resets this event when motor voltage reading is within range.
34	Normal Current Limit Error	S/R	The DCU sets this event when the current limiting circuit does not cap the motor current at the normal current range. The DCU resets this event when the current limiting circuit caps the motor current at the next door opening.
35	High Current Limit Error	S/R	The DCU sets this event when the current limiting circuit does not cap the motor current at the high current range. The DCU resets this event when the current limiting circuit caps the motor current at the next door opening.
36	Current Level Selector Error	S/R	The DCU sets this event when the DCU is unable to activate the normal current limit. The DCU resets this event when the current limits test is executed successfully.
37	Motor OC Shutdown Circuit Failure	S/R	The DCU sets this event when the motor over-current shutdown circuit activation or deactivation time is out of tolerances when tested. The DCU resets this event when the current limits test is executed successfully.
38	Motor OC Shutdown Activated	S/R	The DCU sets this event when motor over-current shutdown limit is reached, meaning that excessive current is used for 2s. The DCU sets this event when the motor over-current shutdown circuit is inactive.
39	5V Logic Supply Out of Range	S/R	The DCU sets this event when the 5V Logic Supply is out of range. The DCU resets this event when the 5V Logic Supply is within range.
40	5V Encoder Supply Out of Range	S/R	The DCU sets this event when the 5V Encoder Supply is out of range. The DCU resets this event when the 5V Encoder Supply is within range.
41	15V Motor Supply Out of Range	S/R	The DCU sets this event when the 15V Motor Supply is out of range. The DCU resets this event when the 15V Motor Supply is within range.
42	V-Gate Open Supply Out of Range	S/R	The DCU sets this event when the V-Gate Open Supply is out of range. The DCU resets this event when the V-Gate Open Supply is within range.
43	V-Gate Close Supply Out of Range	S/R	The DCU sets this event when the V-Gate Close Supply is out of range. The DCU resets this event when the V-Gate Close Supply is within range.
44	Door Closing Visual Light OC	S/R	The DCU sets this event when the feedback signal is inactive with a Door Closing Visual Light turned on. The DCU resets this event when the feedback signal is active with a light turned on.
45	Door Closing Visual Light On w/o Command	S/R	The DCU sets this event when the feedback signal is active without a Door Closing Visual Light turned on. The DCU resets this event when the feedback signal is inactive with a light turned off.
46	Solenoid Low Side OC	S/R	The DCU sets this event when the feedback signal is inactive with a Solenoid Low Side turned on. The DCU resets this event when the feedback signal is active with a Solenoid Low Side turned on.
47	Solenoid Low Side On w/o Command	S/R	The DCU sets this event when the feedback signal is active without a Solenoid Low Side turned on. The DCU resets this event when the feedback signal is inactive with a Solenoid Low Side turned off.

Event #	Event Name	Type	Event Description
48	Spare Output OC	S/R	The DCU sets this event when the feedback signal is inactive with a Spare Output turned on. The DCU resets this event when the feedback signal is active with a Spare Output turned on.
49	Spare Output On w/o Command	S/R	The DCU sets this event when the feedback signal is active without a Spare Output turned on. The DCU resets this event when the feedback signal is inactive with a Spare Output turned off.
50	DCU Temperature Reading Error	S/R	The DCU sets this event when the DCU Board Temperature reading is out of expected range. The DCU resets this event when the reading is within range.
51	Door Open Power TL Stuck	S/R	The DCU sets this event when Door Open (+) TL signal is active for more than 20s. The DCU resets this event when the Door Open (+) TL signal becomes inactive.
52	Door Open Return TL Stuck	S/R	The DCU sets this event when Door Open (-) TL signal is active for more than 20s. The DCU resets this event when the Door Open (-) TL signal becomes inactive.
53	Door Open Crew Key Stuck	S/R	Monitored by L1 and R1 DCU. The DCU sets this event when the Door Open Crew Switch or the Crew Switch Enable signal is activated for more than 20 seconds. The DCU resets this event when the stuck signal is deactivated.
54	Event Log Erased by PTU	P	The DCU logs this event when it receives a request to erase the event log from the PTU.
55	Event Log Retrieved by PTU	P	The DCU logs this event when it receives a request to get the event log from the PTU.
56	-	-	Unused.
57	No Time for More Than 10 seconds	S/R	The DCU sets this event when it has not received the present time after 10 seconds. The DCU resets this event when it receives the present time.
58	System Boot-Up	P	The DCU logs this event after a DCU boot-up.
59	Non-Volatile Data Corrupted	P	The DCU logs this event when a structure in non-volatile memory has been reset to its default value due to CRC16 mismatch at boot-up.
60	-	-	Unused.
61	-	-	Unused.
62	Domain Table Failure Detected by Neuron	S/R	The DCU sets this event when the Neuron Firmware detects a failure. The DCU resets this event when the Neuron Firmware error is cleared.
63	Neuron Application CRC32 Failure	S/R	The DCU sets this event when the CRC32 of the Neuron Application does not match with CRC32 value stored in the Application table. The DCU resets this event when the CRC32 of the application matches with CRC32 value stored in the Application table.
64	-	-	Unused.
65	Door Close TL Stuck	S/R	The DCU sets this event when the Door Close TL is active for more than 20s. The DCU resets this event when the Door Close TL becomes inactive.
66	Door Close Crew Switch Stuck	S/R	Monitored by L1 and R1 DCU. The DCU sets this event when the Door Close Crew Switch signal is active for more than 20s. The DCU resets this event when the Door Close Crew Switch signal becomes inactive.
67	Crew Switch Signal at Invalid Location	S/R	The DCU sets this event when any one of crew switch inputs is active at a non crew switch location. The DCU resets this event when none of the crew switch inputs are active.
68	LSS Mismatch When Unlocked	S/R	The DCU sets this event when there is mismatch between left and right Lock Sensing Switch signals with unlocked door. The DCU resets this event when both Lock Sensing Switches signals are consistent with unlocked door.
69	LSS Mismatch When Locked	S/R	The DCU sets this event when there is mismatch between left and right Lock Sensing Switch signals with locked door. The DCU resets this event when both Lock Sensing Switches signals are consistent with locked door.
70	Crew Switch Open Signals Mismatch	S/R	Monitored by L1 and R1 DCU. The DCU sets this event when the Crew Switch Door Open signal is not consistent with Crew Switch Enable for 8s. The DCU resets this event when Crew Switch Door Open and Crew Switch Enable are consistent.
71	Door Summary Relay High Error	S/R	The DCU sets this event when the DSR High is energized with open and unlocked door for 3s. The DCU resets this event when signals are consistent. ***Note: When DCML is not in normal, the DCU won't set this event.***

Event #	Event Name	Type	Event Description
72	Door Summary Relay Low Error	S/R	The DCU sets this event when the DSR Low is energized with open and unlocked door for 3 seconds. The DCU resets this event when the signals are consistent. ***Note: When DCML is not in normal, the DCU won't set this event.***
73	Zero Speed Carlines Mismatch	S/R	The DCU sets this event when there is mismatch between (+) and (-) Zero Speed CL signals for 2s. The DCU resets this event when both Zero Speed signals are consistent.
74	Crew Switch w/o Zero Speed	S/R	Monitored only by L1 and R1 DCU. The DCU sets this event when any one of Crew Switch signals is active without the presence of Zero Speed signals for 1s. The DCU resets this event when the Zero Speed signals are active or when the Crew Switch signals are inactive.
75	Door Open PWR TL w/o Zero Speed	S/R	The DCU sets this event when the Door Open (+) TL is activated without the presence of Zero Speed signals. The DCU resets this event when the Zero Speed signals are active or when the Door Open (+) TL is inactive.
76	Door Open RTN TL w/o Zero Speed	S/R	The DCU sets this event when the Door Open (-) TL is activated without the presence of Zero Speed signals. The DCU resets this event when the Zero Speed signals are active or when the Door Open (-) TL is inactive.
77	Passenger Door Enable TL Mismatch	S/R	The DCU sets this event when there is a mismatch signals between the (+) and (-) Passenger Door Enable TL signals. The DCU resets this event when both Passenger Door Enable TL signals are same.
78	Passenger Door Enable TL Stuck	S/R	The DCU sets this event when the Passenger Door Enable TL signals are stuck high for more than 20s. The DCU resets this event when both Passenger Door Enable TL signals are inactive.
79	Passenger Door Enable TL w/o Zero Speed	S/R	The DCU sets this event when either the Passenger Door Enable (+) TL or Enable (-) TL is activated without the presence of Zero Speed signals. The DCU resets this event when both Passenger Door Enable TL signals are inactive or both Zero Speed signals are active.
80	Passenger Door Open Pushbutton Stuck	S/R	The DCU sets this event when the Passenger Door Open Pushbutton is stuck high for more than 20s. The DCU resets this event when the Passenger Door Open Pushbutton is inactive.
81	Passenger Door Enable Light OC	S/R	The DCU sets this event when the feedback signal is inactive with a Passenger Door Enable Light turned on. The DCU resets this event when the feedback signal is active with a light turned on.
82	Passenger Door Enable Light On w/o Command	S/R	The DCU sets this event when the feedback signal is active without a Passenger Door Enable Light turned on. The DCU resets this event when the feedback signal is inactive with a light turned off.
83	Door Open TL w/o Mechanical Lock Normal	S/R	The DCU sets this event when the Door Open TL signal is activated without the Mechanical Lock in normal. The DCU resets this event when the Door Open TL is inactive or when the Mechanical Lock is in normal.
84	-	-	Unused.
85	Invalid CPLD Version	S/R	The DCU sets this event when the CPLD returns the invalid version or lower than the minimum expected version. The DCU resets this event when the CPLD returns a version equals or higher than the minimum expected version.
86	CPLD Test Version	S/R	The DCU sets this event when the CPLD returns an unofficial version. The DCU resets this event when the CPLD returns an official release version.
87	Analog Reference Error	S/R	The DCU sets this event when the analog reference value is more than 5% off. The DCU resets this event when the analog reference vale is less than 3% off.
88	Door Closing Audible Alarm OC		The DCU sets this event when the feedback signal is inactive with a Door Closing Audible Alarm turned on. The DCU resets this event when the feedback signal is active with an alarm turned on.
89	Door Closing Audible On w/o Command		The DCU sets this event when the feedback signal is active without a Door Closing Audible Alarm turned on. The DCU resets this event when the feedback signal is inactive with an alarm turned off.
90	No Open Interlock Feedback with Output On		The DCU sets this event when the feedback signal is inactive with an Open Interlock turned on. The DCU resets this event when the feedback signal is active with an Open Interlock turned on.
91	Open Interlock Feedback On w/o Output On		The DCU sets this event when the feedback signal is active without an Open Interlock turned on. The DCU resets this event when the feedback signal is inactive with an Open Interlock turned off.
92	Solenoid High Side OC		The DCU sets this event when the feedback signal is inactive with a Solenoid High Side turned on. The DCU resets this event when the feedback signal is active with a Solenoid High Side turned on.
93	Solenoid High Side ON w/o Command		The DCU sets this event when the feedback signal is active without a Solenoid High Side turned on. The DCU resets this event when the feedback signal is inactive with a Solenoid High Side turned off.
94	Motor Rotation with Door Locked		The DCU sets this event when the motor is able to turn with door locked and closed. The DCU resets this event when the motor does not turn with locked and closed door.

Table A-4: Operating Parameter Description File (Version 0.0) Table

Code	Description	Type of variable	Reading	Sampling
1	Motor Current	Continous	Ampere	50 ms
2	Door Panel Opening Position	Continous	Percentage	50 ms
3	DCU Temperature	Continous	Fahrenheit	50 ms
4	Motor Voltage	Continous	Volt	50 ms
5	Solenoid Lock Switch #1	Logic	True/False	50 ms
6	Solenoid Lock Switch #2	Logic	True/False	50 ms
7	Panel Sensing Switch Left	Logic	True/False	50 ms
8	Panel Sensing Switch Right	Logic	True/False	50 ms
9	Zero Speed PWR CL	Logic	True/False	50 ms
10	Zero Speed RTN CL	Logic	True/False	50 ms
11	Door Open PWR TL	Logic	True/False	50 ms
12	Door Open RTN TL	Logic	True/False	50 ms
13	Door Close TL	Logic	True/False	50 ms
14	Passenger Door Open Pushbutton	Logic	True/False	50 ms
15	Passenger Door Enable PWR TL	Logic	True/False	50 ms
16	Passenger Door Enable RTN TL	Logic	True/False	50 ms
17	Crew Switch Open	Logic	True/False	50 ms
18	Crew Switch Enable	Logic	True/False	50 ms
19	Crew Switch Close	Logic	True/False	50 ms
20	Door Cut-Out Mechanical Lock Normal	Logic	True/False	50 ms
21	Emergency Handle Sense	Logic	True/False	50 ms
22	Door Summary Relay High Side	Logic	True/False	50 ms
23	Door Summary Relay Low Side	Logic	True/False	50 ms
24	Motor Position, Encoder Phase A	Logic	True/False	50 ms
25	Motor Position, Encoder Phase B	Logic	True/False	50 ms
26	Motor Position, Encoder Index	Logic	True/False	50 ms
27	DCU Temperature Sensor	Logic	True/False	50 ms
28	Feedback Door Open Supply	Logic	True/False	50 ms
29	Feedback Solenoid Lock High Side Supply	Logic	True/False	50 ms
30	Feedback Solenoid Lock Low Side Control	Logic	True/False	50 ms
31	Over Current High Limit input	Logic	True/False	50 ms

Code	Description	Type of variable	Reading	Sampling
32	Over Current Normal Limit input	Logic	True/False	50 ms
33	Voltage Monitoring 5V Logic Supply	Logic	True/False	50 ms
34	Voltage Monitoring 5V Encoder Supply	Logic	True/False	50 ms
35	Voltage Monitoring 15V Motor Supply	Logic	True/False	50 ms
36	Voltage Monitoring V-Gate Open Supply	Logic	True/False	50 ms
37	Voltage Monitoring V-Gate Close Supply	Logic	True/False	50 ms
38	Spare Input 1	Logic	True/False	50 ms
39	Spare Input 2	Logic	True/False	50 ms
40	Solenoid Control Activation	Logic	True/False	50 ms
41	Motor Control Activation	Logic	True/False	50 ms
42	Door Closing Visual Alarm	Logic	True/False	50 ms
43	Door Closing Visual Alarm Feedback	Logic	True/False	50 ms
44	Door Closing Audible Alarm	Logic	True/False	50 ms
45	Door Closing Audible Alarm Feedback	Logic	True/False	50 ms
46	Passenger Door Enable Light	Logic	True/False	50 ms
47	Passenger Door Enable Light Feedback	Logic	True/False	50 ms
48	Spare Output Feedback	Logic	True/False	50 ms
49	Running LED	Logic	True/False	50 ms
50	Warning LED	Logic	True/False	50 ms
51	Fault LED	Logic	True/False	50 ms
52	Spare Output	Logic	True/False	50 ms
53	Motor Overcurrent Shutdown	Logic	True/False	50 ms
54	Solenoid Lock Power from CPLD	Logic	True/False	50 ms
55	Open Interlock from CPLD	Logic	True/False	50 ms

Appendix B: Test Report Sheet

(Page 1 of 3)

Item Identification

Item Tested:	
Item P/N:	
Item Version:	

Tester Identification

Tester:	
Time & Date Started:	
Time & Date Finished:	
Physical Location of Test Execution:	

Equipment Used

Description	P/N	Rev	Serial Number	Calibration Due Date
PC running the PTU				
PC Running the PTU simulator				
DCU				

Software Description	P/N	Revision	Version	Date
PTU software				
PTU Simulator Software				

Test Report Sheet (Page 2 of 3)

Comments

Approvals

Tester: _____ Title: _____ Date: _____

Witness: _____ Title: _____ Date: _____

_____ Title: _____ Date: _____

_____ Title: _____ Date: _____

_____ Title: _____ Date: _____

_____ Title: _____ Date: _____

Make Copies of this page and the previous one if required.

Test Report Sheet (Page 3 of 3)

Summary of Anomalies

SPR #	Description